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# A comprehensive review of the current status, developments, and outlooks of heat pipe photovoltaic and photovoltaic/thermal systems

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ABSTRACT

Harvesting energy from the sun is a promising technology to resolve the ongoing energy crisis. Solar photovoltaic (PV) and photovoltaic/thermal (PV/T) systems are mainstream solar energy conversion technologies that enable duple functions to obtain both electrical and thermal energy. However, in practice, emerging problems (uneven temperature distribution, limited cooling effect, freezing issues, high initial costs, etc.) have hindered the development and progress of these technologies. To overcome the aforementioned barriers, a heat pipe was introduced as an effective heat transfer device with passive operation characteristics. The assimilation of cutting-edge thermoelectric (TE), phase change material (PCM) and the development of integrated photovoltaic (BIPV), heat pump, and nanotechnologies have also been explored to achieve higher overall efficiency. This paper focuses on the integration of various heat pipes with solar PV systems and innovative technologies from historical development and recent advancements. In addition, the major observations and challenges are highlighted, and the prospects for future development are corroborated. This work is dedicated to opening new horizons and providing suggestions in this research field, enabling in-depth and broad exploration.

## 1. Introduction

In the context of minimizing carbon emissions and the impact of major environmental contamination, reducing primary energy consumption while preventing harm to the natural environment has received considerable attention. It has prompted a greater emphasis on the utilisation of renewable energy sources. In particular, solar energy has attracted significant attention due to its unlimited and inexhaustible storage capacities and its high compatibility with sustainability [1]. The most prevalent methods of using solar energy are photothermal and photoelectric conversion. These are mature and commercialized methods of gathering solar energy for power and heat.

Solar collectors are specially built heat exchangers that perform photothermal conversion by turning the incoming solar irradiance directly into useable thermal energy.

The gathered thermal energy can be stored in a tank or transported for use through a heat-transfer fluid in a pipeline [2]. Solar photothermal systems are widely used in industrial production and daily life activities, such as agricultural drying, hot water delivery, building heating, solar cooking, and desalination.

Photovoltaic (PV) technology is appealing because the final product is high-grade electrical energy. It is also the most mature solar power-generating technology employed in the commercial sector, with the largest market share of approximately 107 GW in 2020 [3]. This technology is based on the photoelectric effect of a semiconductor material, which uses solar cells as energy-conversion equipment. However, solar cells can only convert a fraction of incoming solar light into electrical energy owing to the limitations of semiconductor materials. The development of the best solar cell conversion efficiency, as shown in Fig. 1, demonstrates that even under laboratory settings, the conversion efficiency of silicon-based solar cells used for non-concentrated conditions is less than 30%, implying that the remaining energy is rejected in vain [4]. Furthermore, there are two negative consequences associated with increasing the operating temperature of solar cells: (a) a drop in the fill factor, maximum output power, and open-circuit voltage of solar cells [5]. (b) Long-term high-temperature operating conditions cause irreversible thermal-stress damage [6].

Therefore, the advanced concept of a photovoltaic/thermal (PV/T) system provides a promising development option that enables the simultaneous collection of heat and electrical energy. The PV module was introduced as a component of the thermal collector, and a thermal absorber plate was used to harvest waste heat from the rear of the PV panel [7]. By driving the working fluid, the generated heat can be

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**Nomenclature**

| | |
|---|---|
| $T_H$ | Hot side temperature, K |
| $T_c$ | Cold side temperature, K |
| $\Delta T$ | Temperature difference, K |
| $\Delta V$ | voltage difference, V |

*Abbreviation*

| | |
|---|---|
| PV | photovoltaic |
| PCM | phase change material |
| TE | thermoelectric |
| PV/T | photovoltaic/thermal |
| BIPV | building integrated photovoltaic |
| TPCT | two-phase closed thermosyphon |
| CTHP | conventional tubular heat pipe |
| LHP | loop heat pipe |
| MHPA | micro heat pipe array |
| PHP | pulsating heat pipe |
| SAHP | solar assisted heat pump |
| DX-SAHP | direct expansion solar assisted heat pump |
| IDX-SAHP | indirect expansion solar assisted heat pump |
| CFD | computational fluid dynamics |
| CPV | concentrating photovoltaic system |
| TEG | thermoelectric generator |
| LCPV/T | low concentrating photovoltaic/thermal |
| HCPV/T | high concentrating photovoltaic/thermal |
| PV/HP-PCM | photovoltaic/heat pipe-phase change material |
| ASHP | air source heat pump |
| SBS | spectrum beam splitter |

removed in time; thus, the temperature of the solar cell can be controlled within an acceptable range. Accordingly, the objective of obtaining electrical and thermal energy is achieved, and address user demand through a single facility.

Over the past years, numerous researchers have focused on enhancing the overall efficiency of solar PV&PV/T systems. Generally, their guiding ideology is the same, which involves regulating the operating temperature of the solar cells and enhance the heat transfer effect. Several research efforts have been directed toward this research theme.

For a solar PV system, cooling demands can be fulfilled by either active or passive cooling. Active cooling technology is the most common project that has received substantial research attention. It uses air and liquid as the cooling medium to extract a portion of the electric energy from the system output (or input from external sources) in order to provide power for dynamic components [8,9]. Specific implementations take a variety of forms, such as water spray [10], mini channel [11], jet impingement [12], forced air cooling [13], nanofluid [14], and two-phase flow [15]. Passive cooling requires no extra energy input, and the temperature of solar cells is controlled by the natural convection of the cooling medium [16], phase change material (PCM) [17], liquid immersion [18], radiative cooling [19], evaporative cooling [20], heat pipe [21], etc.

Solar PV/T systems can be categorised into three major groups based on the type of working fluid: air-based PV/T systems [2], liquid-based PV/T systems [22], and refrigerant-based PV/T systems [23]. Since liquids (i.e., water) have a significantly higher heat capacity and heat transfer coefficient than air, liquid-based systems frequently outperform air-based systems in terms of thermal performance. In addition, owing to refrigerant evaporation in the evaporator, refrigerant-based PV/T systems can also achieve better thermal and electrical efficiencies with an extremely low PV panel temperature. Despite enormous efforts and successes, certain issues still need to be addressed further throughout practical implementations. For instance, active cooling invariably results in a larger temperature difference across the panel and imposes parasitic energy use. Passive cooling has a limited cooling effect, and may be uneconomical. Because of freezing concerns, liquid-based PV/T systems are unsuitable for alpine areas [24]. The high initial cost of the compressor, risk of leakage, and sophisticated design of

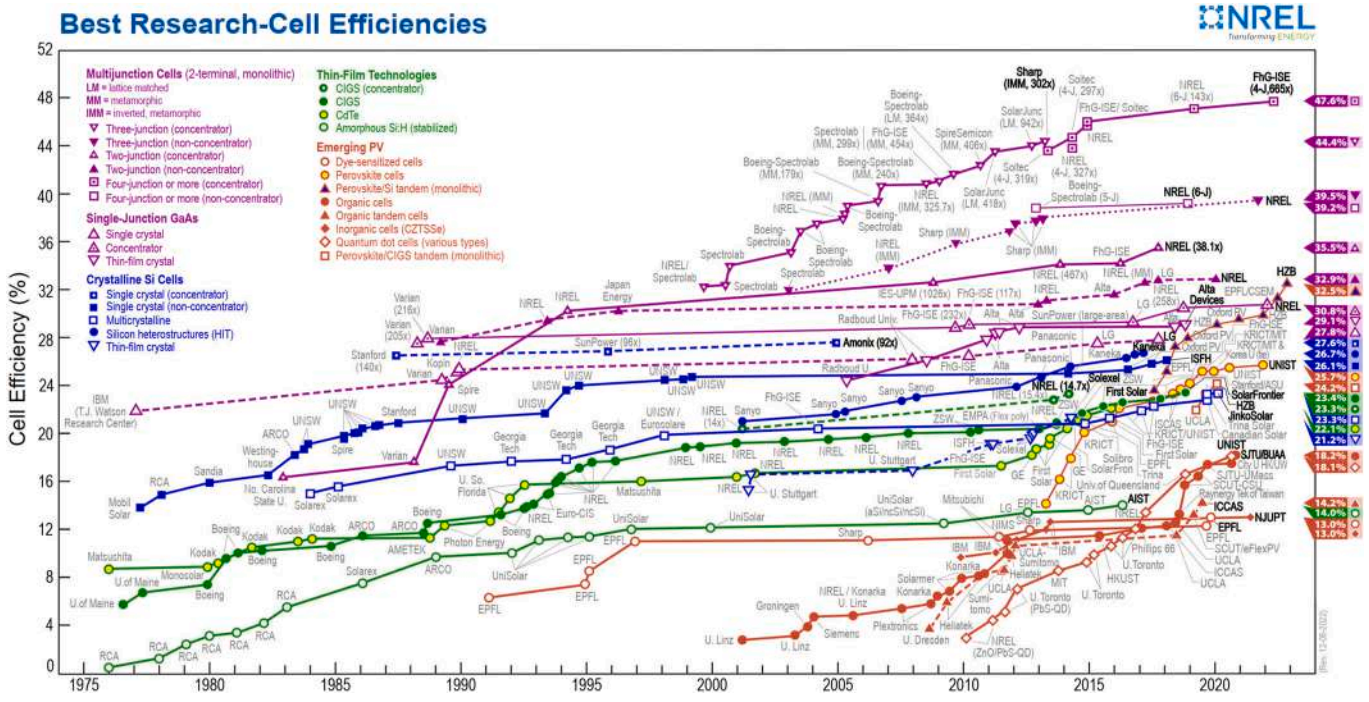



**Fig. 1.** Best conversion efficiencies for solar cells in laboratory conditions (supported by National Renewable Energy Laboratory (NREL), up to 2022) [4].

refrigerant-based PV/T systems need careful evaluation [25,26].

In summary, common PV/T systems (such liquid-based systems) perform well in terms of heat transfer characteristics. However, it proved challenging to produce uniform cooling due to the design of the water channels [27]. Aside from that, ice issues in extremely cold places were inevitable when water was chosen as the working fluid. These drawbacks impede the advancement of conventional PV/T technology.

Hence, some scientific communities have recently turned their attention to heat pipe technology, which has been regarded with the potential to overcome the encountered obstacles. A heat pipe is a highly efficient heat transfer element that is almost isothermal [28]. The non-uniform temperature distribution across the PV panel can be released by integration with a heat pipe; therefore, heat pipe cooling is an alternate and suitable solution for the desire for uniform PV cooling [29]. Heat pipes are automatic equipment, meaning that their operation and maintenance costs are low. The ability of heat pipe cooling systems to achieve simpler construction and economic benefits has been demonstrated by merging heat pipes with PV modules [30]. In addition, by carefully choosing the working medium, the PV/T system based on a heat pipe is fully adaptable to operate in cold conditions without fear of freezing problems [31]. More importantly, to achieve a higher comprehensive energy utilisation efficiency, some popular trends have emerged in recent years, and these trends combine the expansion of heat pipe PV&PC/T systems with other innovative technologies. Some of these tentative exploratory efforts include: Thermoelectric (TE) technology [32], PCM technology [33], heat pump technology [34], and building integrated photovoltaic (BIPV) technology [35], and nanotechnology [36].

Based on the preceding literature review, using heat pipes in solar energy systems, including solar PV and PV/T systems, is a possible solution for addressing the issues experienced in normal systems. To date, some phased summaries have been published regarding the use of heat pipes in solar PV or PV/T systems. For example, a review study conducted by Zhou et al. [37] summarized the structure and operational principles of the heat pipe PV/T system, and pointed out the research gaps and future trends. Nonetheless, previous studies do not provide a full study and collection of current heat pipe PV&PV/T systems. Despite a slew of articles on the use of heat pipes in PV cooling or PV/T collectors, the majority of them failed to give specific advancements in integrated PV&PV/T systems combined with innovative technologies [5,25,31,38,39]. The purpose of the current paper is to give references for readers and the scientific community, as well as to respond to pertinent queries about heat pipe PV&PV/T systems. PV& PV/T systems based on heat pipes offer various possibilities and paths for overcoming the aforementioned challenges. Hence, the answers to the following inquiries demonstrate the significance and novelty of this review study.

- What exactly are heat-pipe PV&PV/T systems? What about the operating principle of the heat pipe PV&PV/T systems? Which types of heat pipes can be used in these systems? How can heat pipes be combined into these systems?
- What kinds of novel technologies can be used in heat-pipe PV&PV/T systems? What about their operating principles? How far has this research progressed?
- Finally, what are the challenges in the current research? What are the future prospects?

Therefore, this review presents comprehensive results and discussions regarding the use of heat pipes in PV&PV/T systems combined with innovative technologies. A schematic view of the corresponding heat pipe PV&PV/T system is illustrated in Fig. 2. To provide a better grasp of the relevant issue, the structure of the current article is shown in Fig. 3. The fundamentals, operation principles, and classifications of the PV, PV/T, and various heat pipes are summarized in Section 2. Section 3 presents the development and status of research studies involving sole PV or PV/T systems. In Section 4, a summary of the progress considering the combination of innovative technologies with heat pipe PV&PV/T systems is analyzed, and the last section outlines the main conclusions, challenges, and outlook for future studies.

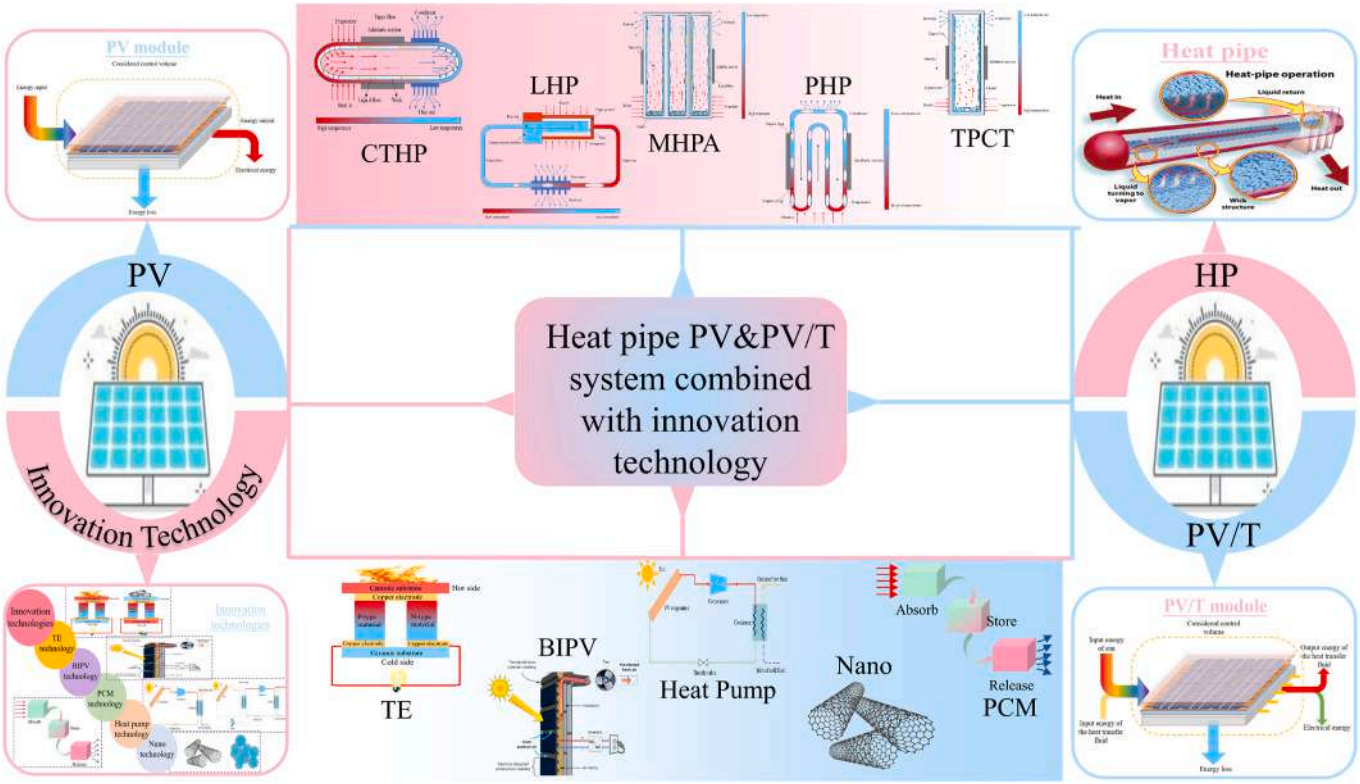



**Fig. 2.** Schematic of the heat pipe PV&PV/T systems combined with innovative technology.

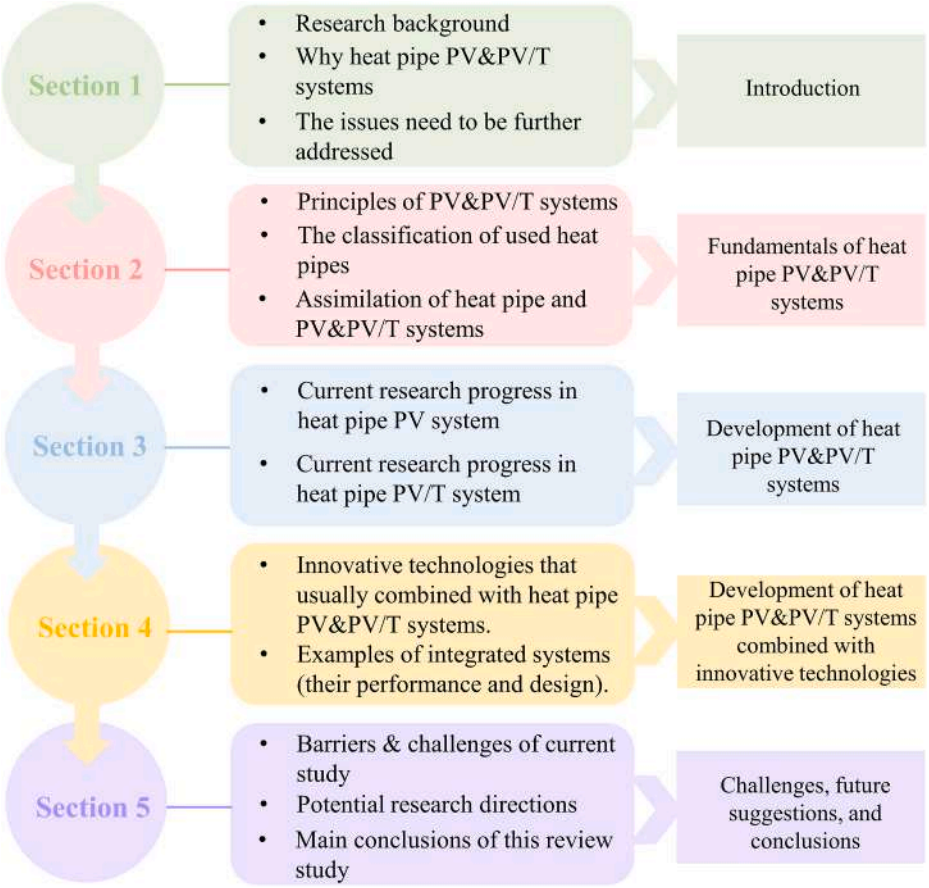



**Fig. 3.** The structure of the current work.

## 2. Fundamentals of heat pipe PV&PV/T systems

### 2.1. Principles of PV&PV/T modules

Nearly two centuries have passed since physicist Becquerel discovered the photovoltaic effect. The PV power generation technology has advanced and developed significantly throughout this period. Although various cell types (e.g., CdTe, GaAs, Mono-*c*-silicon, Poly-*c*-silicon) have been proposed, designed, and fabricated to boost the energy conversion efficiency of solar cells, silicon-based solar cells remain the most successful and widely used technology, as shown in Fig. 4. A typical structure of this type of solar cell is depicted in Fig. 5a. When sunlight strikes the surface of a material, photons excite the electrons in the covalent link, creating electron-hole pairs inside the substance. Subsequently, the electrons and holes in the two types of materials migrate to the two ends of the material owing to the influence of the internal electric field, and a current is then generated [40,41].

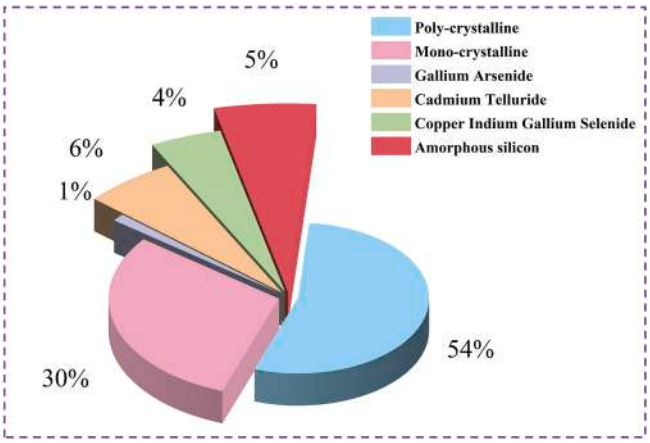



**Fig. 4.** Market share of different types of solar cells [41].

The power generated by a single solar cell is relatively small and falls short of satisfying actual use requirements. Consequently, connecting a set number of solar cells in parallel or hybrid connections is common practice to significantly increase the total output power. Furthermore, solar cells are relatively brittle and cannot be exposed to the environment for extended periods of time. Hence, TPT material, a polymer compound with the functions of sealing, insulation, and waterproofing, has been utilised to prevent the external environment from invading the solar cells. EVA, a different polymer compound, was employed to attach and secure solar cells to tempered glass and TPT back sheets. Subsequently, various layers were bonded to an aluminium frame [42]. The detailed construction of the commercial PV panel is shown in Fig. 5b. A typical solar PV generation system was established with the assistance of other auxiliary equipment (charge controller, battery, and inverter).

In general, solar cells can effectively use only a small portion of the received solar energy, with the remainder being converted into heat energy. To achieve a higher energy conversion efficiency, it is highly recommended to combine a PV module with a solar collector in a hybrid system [31]. In 1976, Wolf examined the notion of a hybrid PV/T system for single-family dwellings [43]. Later, Kern and Russell described a similar system that used water or air as transfer fluids [44]. This fluid could remove the extra heat rejected by the PV module, allowing the temperature of the solar cells to be maintained correctly. Moreover, the total efficiency of solar photovoltaic/photothermal use per unit sun-receiving area was enhanced [45]. Compared with single solar water heaters or solar PV systems, the dual functionalities of the

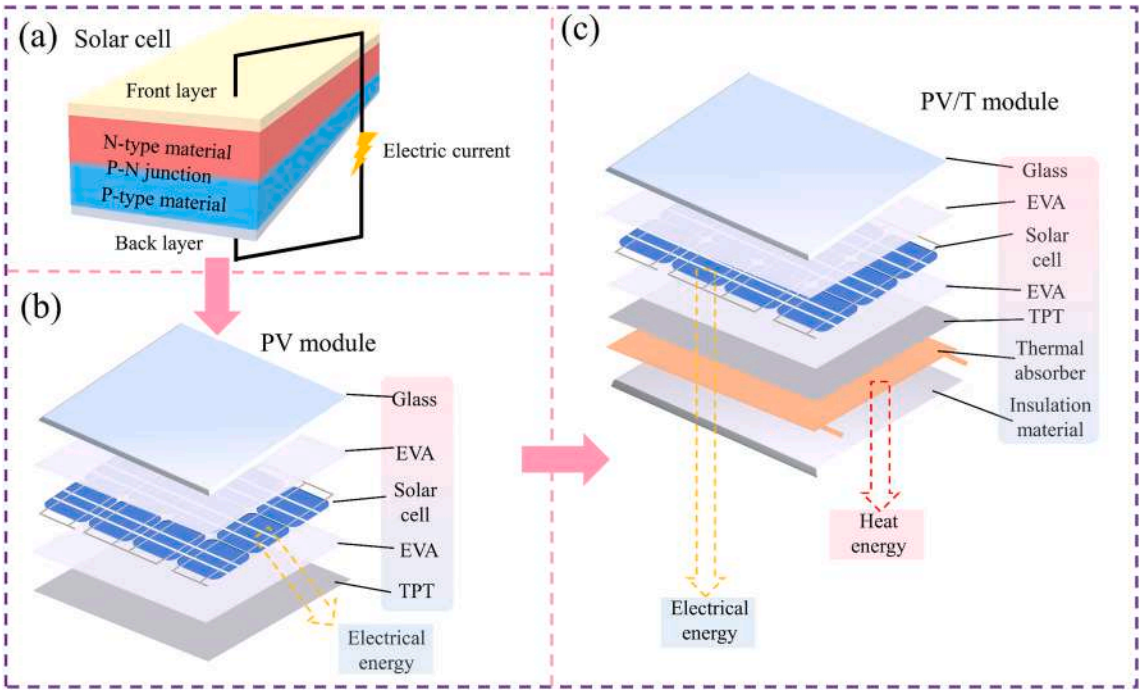



**Fig. 5.** Typical structure of (a) silicon-based solar cell, (b) PV module, and (c) PV/T module.

proposed PV/T system can achieve greater total solar energy utilisation efficiency [46,47].

The most popular method for implementing a PV/T module is to mechanically or chemically attach a commercial solar PV module to a thermal absorption panel. Thermal absorber plates can be used in several ways (for example, sheet-and-cylindrical tube structures, sheet-and-semi-round tube structures, flat-plate tube structures, and heat pipe structures.). Fig. 5c shows the typical layout and composition of the PV/T collector.

### 2.2. The classification of used heat pipes

Heat pipe technology has attracted considerable interest as a passive heat transfer solution owing to its ability to carry enormous amounts of

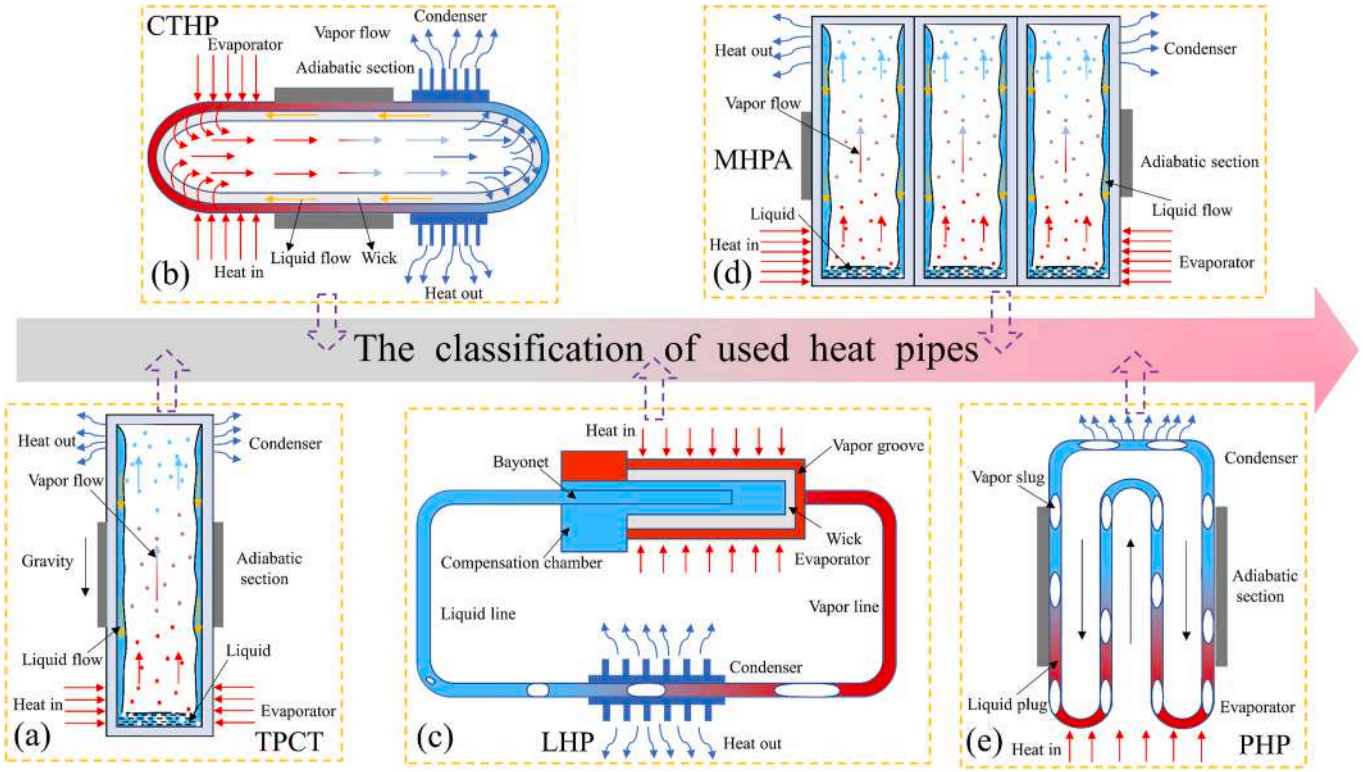



**Fig. 6.** Operation principles of the used heat pipes (a) TPCT, (b) CTHP, (c) LHP, (d) MHPA, and (e) PHP.

heat over long distances without utilizing any moving equipment [48]. Owing to their outstanding transfer capabilities, high heat transfer efficiency, and strong isothermal properties, they can be used to solve heat transfer problems in thermal engineering applications [49]. The term "heat pipe" refers to a broad notion that may be used to characterise several heat transfer devices that adopt a gas-liquid two-phase flow heat transfer mechanism. The only distinction is the driving power, which determines how the condensate liquid returns to the evaporator portion (e.g., gravity force, capillary force, centrifugal force, surface tension, and pulse force). To gain a better understanding, the following section provides a brief overview of the heat pipes used in solar PV and PV/T systems in terms of categorisation and operation. Heat pipes commonly used in solar energy systems are classified into five types based on their structure: two-phase closed thermosyphon (TPCT), conventional tubular heat pipe (CTHP), loop heat pipe (LHP), micro heat pipe array (MHPA) (or microchannel heat pipe, MCHP), and pulsating heat pipe (PHP).

#### 2.2.1. Two-phase closed thermosyphon (TPCT)

The TPCT was a gravity-assisted heat pipe. The key difference between TPCT and traditional heat pipes is that a wick structure is absent inside the tube. The TPCT comprises three parts: an evaporator, an adiabatic section, and a condenser section that creates a vacuum within the sealed pipe, partially filling it with heat transfer fluid [36]. Fig. 6a depicts the basic structure and operation of the TPCT. According to this schematic, the condensate liquid returning from the condensation section to the evaporation section was propelled by gravity rather than by the capillary force provided by the wick structure. The gravity-assisted aspect resulted in a TPCT with completely distinct properties. First, the tilt angle between the heat pipe and ground normal has a significant impact on the TPCT operating performance. Second, the TPCT condensation section must be located above the evaporation section to ensure that the condensate liquid returns to the evaporator. The TPCT can be developed as a thermal diode that acts as a thermal switch based on the one-way heat transmission feature.

#### 2.2.2. Conventional tubular heat pipe (CTHP)

A conventional tubular heat pipe (CTHP) is a round-shaped heat pipe. Initially, Gaugler outlined the circular heat pipe concept in a patent filed in 1942, and Grover later refined it in the early 1960s. It was not until Grover's contribution in elucidating CTHP's exceptional heat transmission capabilities of the CTHP that the heat pipe was recognised and developed further [50]. The wick structure used in the inner wall of the tube is the most obvious distinction between the CTHP and the other types of heat pipes. In the case of the CTHP, the wick structure allowed it to defy the pull of gravity. The condensate liquid could flow back to the evaporator section with the help of the capillary force provided by the wick, regardless of the heat pipe orientation. The gravity force enhances the capillary force if the CTHP is arranged parallel to the direction of gravity (the evaporator portion sits at the lowest position). Fig. 6b shows the basic structure of the CTHP and its operating mechanism. The capillary force created by the wick was critical for the CTHP function, which increases the thickness of the wick, implying that heat may be transported over a longer distance. However, increasing the thickness also leads to an increase in the radial heat transfer resistance, which degrades the heat transfer capacity of the heat pipe. Moreover, while selecting the wick structure and materials, the characteristics of the working fluid must be examined to avoid any chemical interactions that may occur between the heat transfer medium and wick material [50]. The geometry of the wick can be portrayed in various ways, with meshes, sintered wicks, and grooves being the most frequent [51].

#### 2.2.3. Loop heat pipe (LHP)

The LHP is a kind of heat pipe that can effectively address the inherent shortcomings of the CTHP in long-distance heat transmission and the direction of cold and heat sources, as shown in Fig. 6c. It is an appealing heat-transfer device with a high heat-transfer capacity over extended lengths of dozens of meters, which is difficult to achieve using traditional heat pipes. In addition to the evaporator and condenser sections observed in the CTHP, the LHP has an evaporator and condenser section. The evaporator is the heart of the LHP and performs two critical functions: collecting heat from a heat source and producing power for the circulation process. More specifically, the compensation chamber (CC), wick, and bayonet constitute the primary components of the evaporator body (liquid guide tube). The vapour groove was the axial channel outside the wick, and the main liquid line was the interior channel of the wick. CC is a unique LHP structure that is utilised to ensure liquid storage and supply to the wick of the evaporator during LHP operation, and to confine non-condensable gas and prevent air lock. Another distinction between LHP and traditional heat pipes is the separation of vapour and liquid lines. As vapour and liquid are transferred into separate pipelines, entrainment is prevented. When the evaporator is subjected to an external heating load, the internal heat transfer working medium is heated and evaporated on the outer surface of the wick. Consequently, the created vapour flows out of the vapour channel into the vapour line, and then into the condenser to complete the condensation. The latter liquid entered the liquid guide tube through the liquid line to replace the evaporator wick, resulting in a revolving cycle.

#### 2.2.4. Micro heat pipe array (MHPA)

Initially, Cotter discussed the potential and principles of micro-heat pipes at the 5th International Heat Pipe Conference in 1984. The researchers suggested the concept of heat pipe miniaturisation. They defined a micro heat pipe as having a mean curvature of the vapour–liquid interface equivalent in size to the reciprocal of the hydraulic radius of the inner flow channel [52]. The heat transmission capacity of a single micro heat pipe is restricted owing to the micro-scale features. However, this vulnerability can be solved by joining numerous tiny single heat pipes in parallel to form an array. As shown in Fig. 6d, MHPA is a unique heat transfer element that satisfies the requirements for high-intensity heat transmission. The interior of the MHPA consists of numerous separate microchannels, each of which is a self-contained, micro-heat pipe. Owing to the extra specific construction required for the closest contact with the heat exchange surface, numerous contact thermal resistances, cylindrical form constraints, and diminishing heat transfer contact surfaces are unavoidable for the CTHP. However, in the case of the MHPA, the precisely constructed plate structure improved the form flexibility and heat transfer capability of the target, which required cooling. In addition, the presence of an inner micro-fin structure in each micro heat pipe significantly enhances the heat transfer area. Furthermore, heat pipes are often constructed from metals with high thermal conductivities. Therefore, the wall construction between the two micro heat pipes reinforces the supporting structure and conducts heat from one side to the other, causing a phase change transformation to occur on the entire wall surface of the micro heat pipe. Because the MHPA is a collection of micro heat pipes, its functioning concept is almost identical to that of the TPCP. In an MHPA, micro heat pipes have a connection that is both autonomous and highly united. This independence implies that the operation of each heat pipe does not interfere with the operation of the other heat pipes; for instance, if one micro heat pipe breaks down, it does not affect the operation of the other heat pipes.

#### 2.2.5. Pulsing heat pipe (PHP)

In 1990, the Japanese researcher Akachi proposed a PHP with a revolutionary construction [53]. PHP is divided into two types: closed-loop and open-loop PHP. The ends of both the tubes were not joined in an open-loop PHP. In contrast, a closed-loop PHP is formed by connecting both ends of the tubes to produce a closed system [54]. Fig. 6e illustrates the main construction and operation of the closed-loop. A PHP is a heat transfer device with long and tiny inner diameter tubes that have been bent into serpentine forms. The interior of the capillary tubes was evacuated and partially filled with the working

fluid [55]. The driving force of the circulation inside the PHP is the continuous growth and collapse of bubbles in the tubes, resulting in the pulsating motion of the fluid, which is not produced by the capillary force in the wick structure of conventional heat pipes [53]. It may be further stated as follows: in its initial condition, the vapour slug and liquid plug of the PHP were separated and dispersed randomly. The heat transfer fluid then absorbs heat and creates bubbles, which rapidly expand and pressurise, causing the heat transfer fluid to flow into the condenser. However, owing to the pressure differential between the evaporator and condenser, as well as the imbalanced pressure of nearby tubes, the heat transfer fluid oscillates between the evaporator and condenser, which completes the heat transfer process.

Therefore, the simple low-cost structure of a PHP, ease of manufacture due to the lack of wick structure, compact size and small space occupation, quick start up, and high heat flux adaptability are excellent characteristics that make it an alternative scope to the effective thermal management and heat transfer device [55]. However, the theoretical mechanism behind the complex physical phenomena is still unclear because the different gas-liquid two-phase flow and heat transfer processes in PHP are intricate; therefore, an in-depth investigation is required.

### 2.3. Assimilation of heat pipes and PV&PV/T systems

The heat pipe PV and PV/T systems are a hybrid of the previously mentioned heat pipes and PV&PV/T configurations. A schematic of the heat-pipe PV system is shown in Fig. 7a. In this layout, a heat pipe was chosen as the heat sink for the PV module. Typically, the evaporator side of the heat pipe is joined to the PV panel by a thermally conductive adhesive, while the condenser part is directly exposed to environmental conditions. The heat pipe absorbs waste heat from the PV module and releases thermal energy at the condenser. Thus, the operating temperature of the PV module was maintained within permissible ranges. In the heat pipe PV/T system, the evaporator side was also attached behind the PV panel, and the rejected heat from the condenser section was collected by the heat transfer fluid, as illustrated in Fig. 7b. The heat is continually collected by circulating the coolant through the pipeline, and the accumulated heat can be stored for a hot water supply, room heating, or other purposes. In this process, insulation materials are used to minimize the heat exchange between the system and environment to reduce the heat loss from the collector.

## 3. Development of heat pipe PV&PV/T systems

The utilisation of heat pipes in solar PV and PV/T systems is a viable method for controlling the increasing operating temperature of solar cells, particularly where uniform cooling is required. Over the past decades, researchers have explored and tested various types of heat pipe PV&PV/T systems, both numerically and experimentally. Efforts have been made to find an optimal design and layout to achieve higher energy conversion efficiency. This section discusses the development and progress of these aspects, as well as the most recent research that has appeared in recent years.

### 3.1. Using heat pipes in PV system

As a general rule, solar PV power generation systems can be categorised into non-concentrating PV systems and concentrating PV systems. A non-concentrating PV system is the most fundamental way to harness solar energy because it does not require any extra optical concentrators or a sun-tracking mechanism. Many noteworthy studies have been conducted in this field to enhance the performance of solar cells. The representative studies are described in the following sections.

To enhance the heat transfer effect of the heat pipe, two computational fluid dynamics (CFD) models of finned heat pipes were developed and validated by Hughes et al. [56] and Koundinya et al. [57] to remove heat from the rear of the PV panel. They confirmed that using a finned heat pipe is a promising method to control the operating temperature of PV panels near the ambient temperature. As illustrated in Fig. 8a and **b**, a prototype of a nanocoated flat-plate heat pipe was modified by Du and Zhang et al. [30,58,59] to cool down the solar cell. A piece of a bare solar cell without encapsulation was placed on the evaporator side of the heat pipe. Subsequently, the gap between the cell and heat pipe was filled with heat conduction glue to reduce the thermal barrier. Different heat sinks (passive fins and an active water box) were attached to the condenser to investigate the cooling effect. Based on their findings, the temperature of the solar cell could be significantly reduced using a heat pipe (as shown in Fig. 8c). In addition, the temperature distribution across the solar cell was evenly compared to the reference solar cell, and the temperature differences in the x and y directions of the heat-pipe-cooled solar cell were approximately 1 K and 2.5 K, respectively, as shown in Fig. 8d.

Alizadeh et al. [60,61] numerically investigated a PHP-based solar PV cooling system, as displayed in Fig. 9a and **b**. They compared the performance of a PHP-cooled PV system with other cooling strategies (copper heat sink, natural air convection, and active water-forced convection). The results revealed that employing PHP as passive cooling augments yield in an 18% increase in electrical power and a 10% shorter payback period when compared to an active water-cooled PV system. Three-dimensional temperature diagrams of the systems are presented

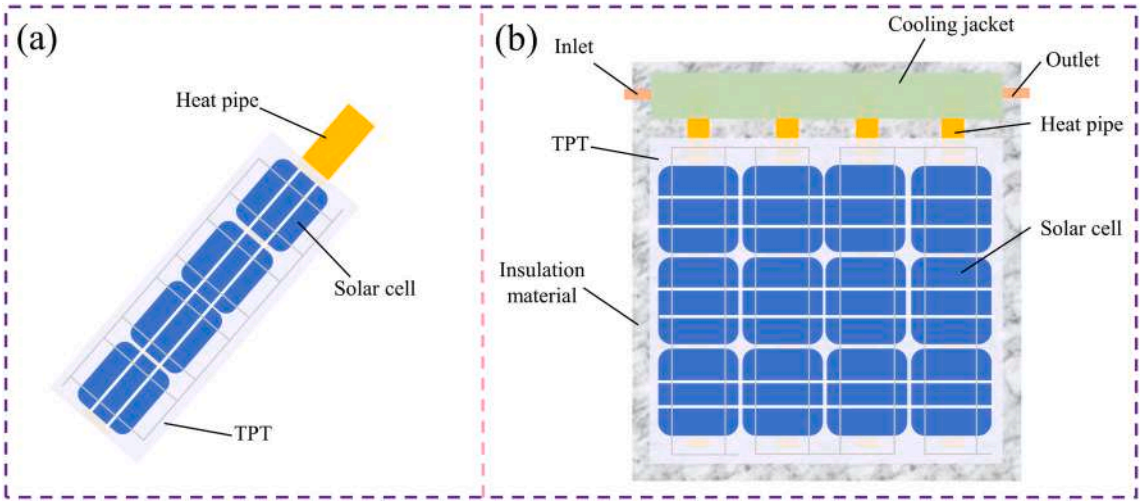



**Fig. 7.** Hybrid layouts of (a) heat pipe PV module, and (b) heat pipe PV/T module.

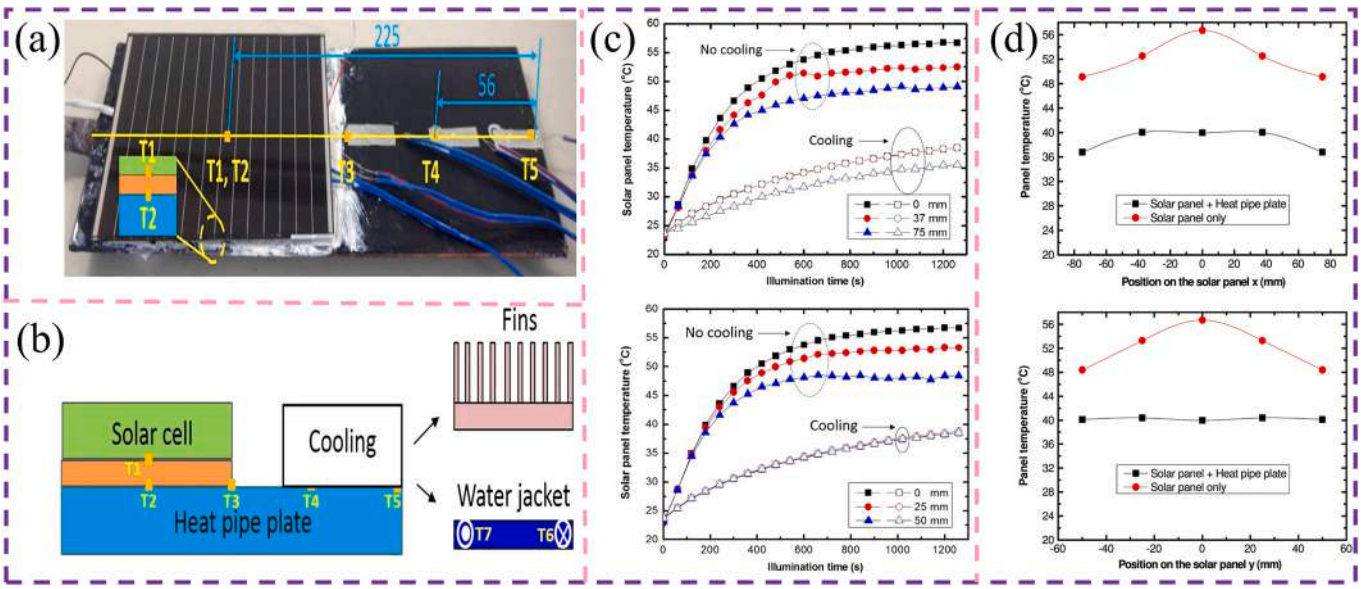



**Fig. 8.** Nano-coated flat plate heat pipe for cooling solar cell, (a) actual setup, (b) schematic, (c) variation of solar cell temperature under 600 W/m$^2$, (d) temperature distributions of solar cell under 600 W/m$^2$ (adapted/reproduced with permission from Ref. [58]). Copyright © 2016 John Wiley and Sons.

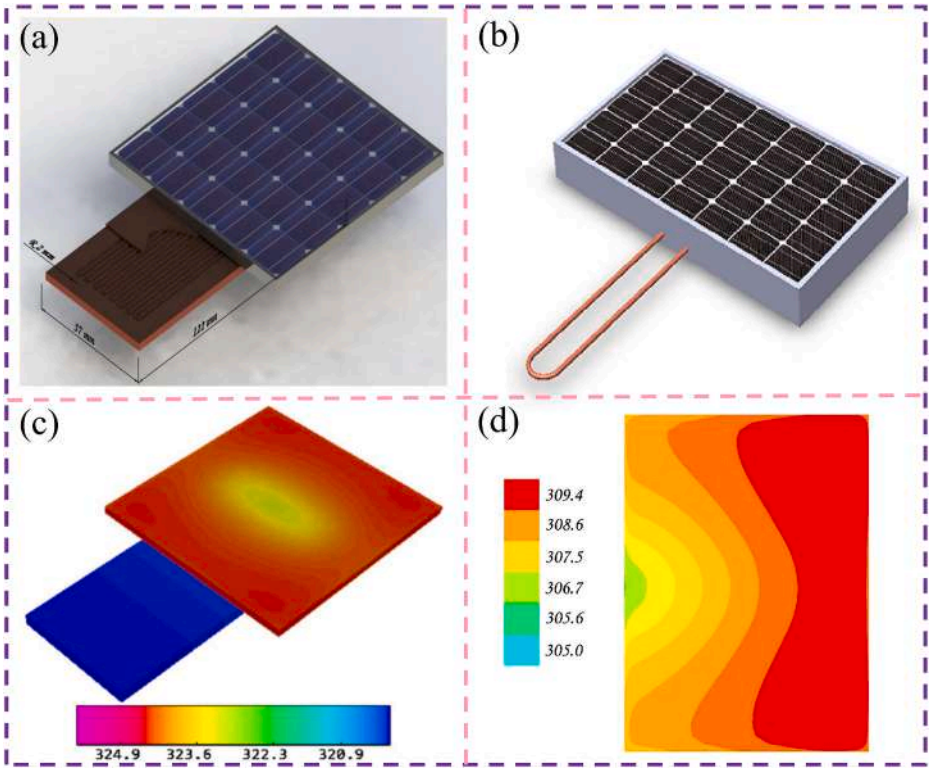



**Fig. 9.** Cooling using PHP (a) multi-turns of PHP, (b) single turn of PHP, (c) temperature distribution of PV panel cooled by multi-turns PHP, and (d) temperature distribution of PV panel cooled by single turn PHP (adapted/reproduced with permission from Refs. [60,61]). Copyright © 2020 and 2018 Elsevier.

in Fig. 9c and **d**.

Sutanto et al. [62] investigated the transient performance of a floating PV system cooled by a loop TPCT and found that the operating temperature of the PV panel could be effectively decreased. The use of heat pipes in floating PV systems is highly recommended because natural water sources can be used directly as the cooling medium of the heat pipe. Besides, no power equipment was required in this design. Aamir et al. [63] performed an optimum investigation that accounted for the pitch of each heat pipe. The obtained data revealed that the plan with 50 mm spacing had a greater temperature decrease of 9 K. In addition, Al-Amri et al. [18] attempted to achieve a uniform PV surface temperature by combining heat-pipe cooling and liquid-immersion cooling under harsh environmental conditions. They discovered that a temperature drops of 53% was obtained when compared to the reference arrangement.

The low energy density is the primary drawback of solar energy. If the incoming light can be concentrated to improve the radiation density of the solar receiver, the PV system may employ less expensive cell

materials. Adopting concentrating technology to overcome this constraint is a popular option. However, the increased input heat flux intensity in the concentration condition suggests a massive heat build-up, which poses an acute problem for the thermal management of solar cells. Hence, the next part will consider the use of heat pipes in PV systems under concentrated conditions.

The concept of using a heat pipe in a concentrating photovoltaic (CPV) system could be traced back to Russell's study [64] in 1982. As illustrated in Fig. 10, numerous PV cells were installed linearly on the surface of the heat pipe, and each was fitted with a linear Fresnel lens to focus the solar radiation on the solar cells. The pipeline inserted into the vapour chamber can deliver sufficient cold water to condense internal steam. This design presented a complete method to cool down solar cells under concentrated conditions, and inspired later researchers to perform similar studies.

To enhances the heat transfer effect, many researchers have suggested the use of finned heat dissipators to provide a more intense condensation effect for efficient heat pipe operation. Akbarzadeh and Wadowski [65] studied the performance of a solar cell cooled by TPCT under 20X suns. The finned heat sink was laminated to the condenser side of the heat pipe to expand the heat-dissipation area. The results revealed that the temperatures of the solar cell under cooled and uncooled conditions were 319.15 K and 357.15 K, respectively, which resulted in a 10% increment in efficiency. A simple-to-process finned--heat pipe device was created by Emam et al. [66] to cool the single-junction solar cells. The experiments were carried out using a variety of working fluids (water, ethanol, methanol, and an ethanol-based water solution) and charging ratios. They claimed that the increase in electrical power was between 4 and 8.18%. Cheknane et al. [67] designed a matrix heat sink to cool down a CPV system exposed to 500X suns using water and acetone as coolants. The most significant contribution of their study was the demonstration that the use of passive heat-pipe management techniques is feasible for cooling solar cell under 500X concentrated solar radiation conditions. Similarly, Kuo et al. [68] designed a modular solar engine coupled with a solar cell and heat pipe under ultrahigh concentration ratio conditions. The system employed triple junction III-V cells with an effective area of 0.36 $cm^2$ as a receiving chip. The authors believed that the use of solar engines is beneficial for CPV systems owing to their low operation and maintenance costs. According to their test, the coupled system could easily sustain solar radiation of more than 800X suns and was easy to manufacture. Anderson et al. [69,70] conducted a series of experiments and simulations to determine the ideal arrangement of an Al plate fin design affixed to the surface of a heat pipe. Water was used as the working medium in the copper heat pipe. The size and spacing of the Al fins under air convection were carefully designed by the authors using numerical methods. The optimum temperature difference between the solar cell and the environment was found to be approximately 313.15 K, indicating strong thermal regulation capability. Li et al. [71] attempted to overcome the non-uniformity of the temperature distribution of a GaAs solar cell using a heat pipe. They designed four different structures

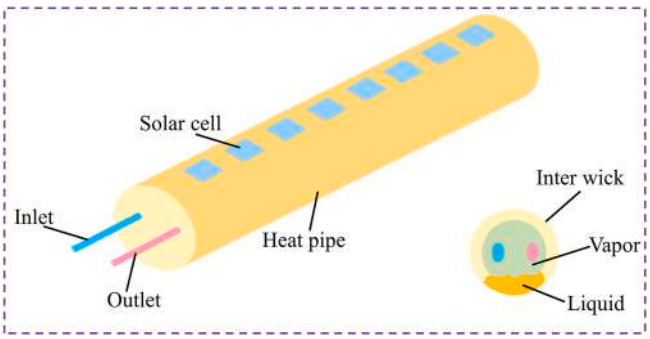



**Fig. 10.** Cooling of concentrating solar cells using heat pipe [64].

equipped with a flat plate heat sink, rectangular fins, small fins, U-type or L-type heat pipes. The simulation results revealed that that the system equipped with U-type or L-type heat pipes possessed a minimal temperature difference of 2 K and 3 K between the central and peripheral regions. In other words, the use of a heat pipe resulted in a better temperature uniformity effect and aided in thermal management. On-site testing of a CPV system adopting a hybrid structure heat pipe was performed by Huang et al. [72], as shown in Fig. 11a. In this study, heat pipes with sintered material structures, mesh screen structures, supporting structures, and no supporting structures were experimentally examined. The thermal resistances of the supporting structure and sintered structure were 72% and 65% less than the no supporting structure and mesh screen structure. The results also revealed that the total electrical efficiency of a solar cell cooled by a heat pipe with a sintered capillary was greater than that of a system with a normal copper mesh screen construction. A two-stage dish CPV system using a triple-junction solar cell cooled by a finned heat pipe was analytically and experimentally studied by Wang et al. [73], as shown in Fig. 11b. The results show that the average generated electricity and electrical efficiency were 1.52 $W/cm^2$ and 29.3%, respectively, under an input solar irradiance of 450 $W/m^2$.

Armijo et al. [76] examined the performance of single- and dual-junction solar cells cooled using a TPCT device. The results revealed that both single- and dual-junction solar cells exhibited higher performance (improvements of 1.7% and 2.1% in electrical efficiency) under sun concentrations of 132X and 560X, respectively, when compared to the reference system. Moon et al. [77] numerically studied a heat-pipe-based CPV system. Based on their findings, the temperature difference between the cooled and uncooled states of a solar cell was approximately 32 K. The influence of various operating fluids in the loop TPCT for CPV cell cooling was numerically investigated by Chen et al. [74] (shown in Fig. 11c). An electric heating sheet was employed to simulate the heat produced by the solar cell, and an insert of the graphene sheet was placed between the solar cell and the evaporator part of the loop TPCT. The numerical data indicated that the coolant of acetone could be applied for cooling the solar cell under 7X suns, whereas water and ethanol were suitable for the 1X and 3X conditions, respectively. Wang et al. [78] developed a unique hybrid CPV system that could operate under various cooling conditions. The passive condition employed numerous heat pipes as heat spreaders, whereas the active condition employed water to remove heat. When the heat load was less than 13 $W/cm^2$, the findings revealed that the passive condition was more competitive. Wang et al. [75] developed a novel form of TPCT at atmospheric pressure for cooling low-concentration PV cells, as shown in Fig. 11d. According to the authors, this unique technique exhibited good temperature uniformity. The largest temperature difference was approximately 5.6 K when the solar radiation was approximately 3000 $W/m^2$. As shown in Fig. 12, Anand et al. [79] fabricated a solar PV system cooled by an LHP and tested its performance in outdoor conditions. The measured temperature differences between the solar cell and the environment were 310.95 K, 327.31 K, and 345.75 K for the heat pipe cooled system, heat sink system, and solo PV systems, respectively. Soliman et al. [80] reported on the use of flat heat pipes for solar cell thermal management. At a solar radiation energy of 3000 $W/m^2$, the acquired results suggest that the power production may be increased by 9.1% under the heat pipe cooled condition compared to the uncooled mode.

The possibility of adopting a PHP to cool down the CPV system was examined by Geng et al. [81,82]. The results revealed that the PHP device was suitable for cooling a CPV cell under 12X suns and the output power could be significantly enhanced. Similarly, Raj et al. [83] and Wang et al. [84] presented CPV systems cooled by PHP using a mirror and a dish concentrator, respectively. Both results confirmed that the PHP can be considered as an alternative heat sink for cooling solar cells under concentrated conditions. The dish CPV system using the 3D PHP as a heat sink is illustrated in Fig. 13.

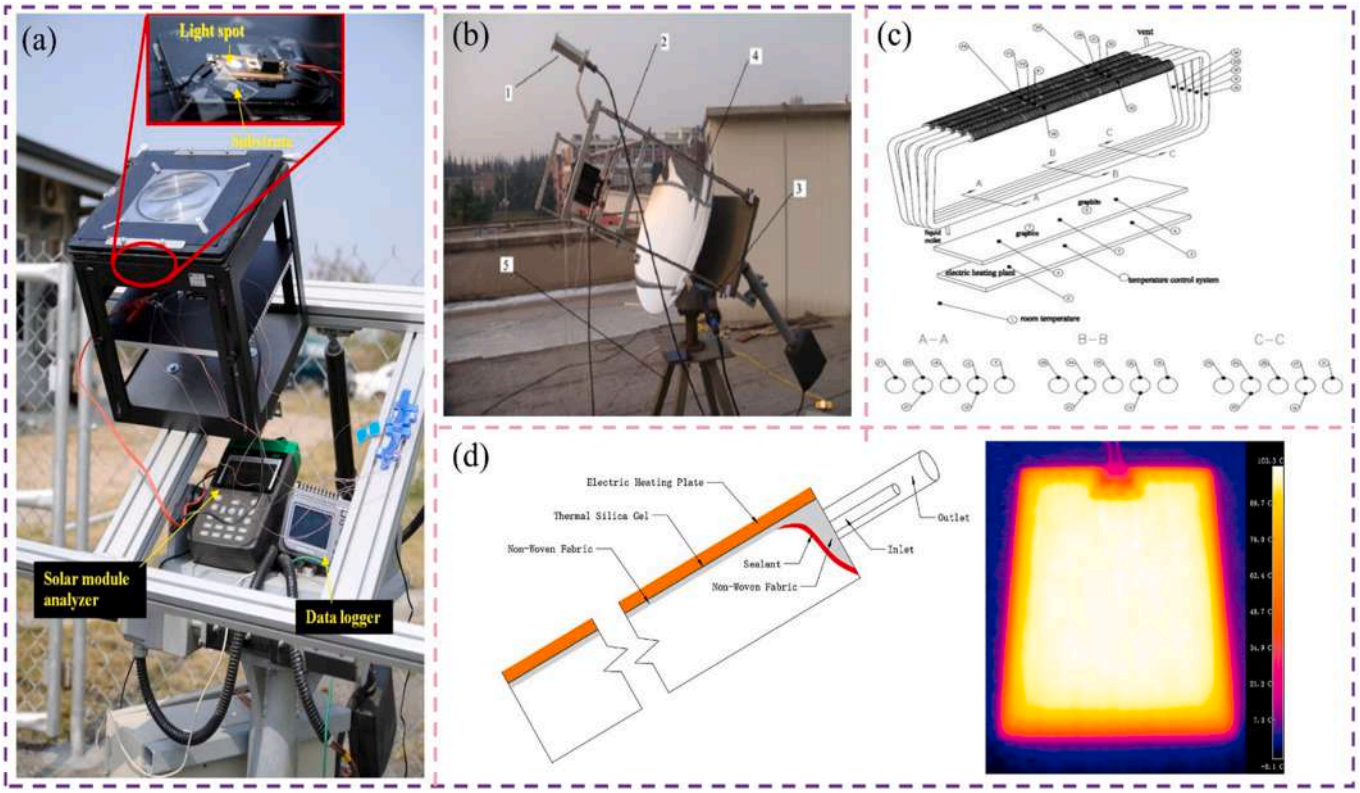



**Fig. 11.** Images of typical CPV system (a) tested hybrid structure heat pipe CPV system [72]. (b) a two-stage CPV system (adapted/reproduced with permission from Ref. [73]). Copyright © 2013 Elsevier. (c) setup of cooling system using loop TPCT (adapted/reproduced with permission from Ref. [74]). Copyright © 2016 Elsevier. (d) schematic view of the atmospheric plate thermosyphon (adapted/reproduced with permission from Ref. [75]). Copyright © 2018 Elsevier.

### 3.2. Using heat pipes in PV/T system

Utilizing heat pipes in a PV/T system not only improves the electrical performance of the PV panel but also allows more energy per unit area compared to a pure PV system or a solar thermal collector. This section describes the major works of the heat-pipe PV/T system. To date, compared with other types of PV/T systems (e.g., air type or water type), the number of research samples in the heat-pipe PV/T system is far from sufficient, and research development is still in the initial stage. Fortunately, there have been some preliminary studies on prototypes as references for follow-up researchers. For example, a classical investigation by Moradgholi et al. [85] focused on the effects of various working fluids. A TPCT device was used to construct the PV/T system, and its performance was examined during the summer and spring seasons (Fig. 14a). Accordingly, two water jackets connected in series were installed at the head of the PV module to provide cooling capacity, and the condenser of the TPCT was directly immersed in the jackets. The system tested in the summer condition was charged with acetone with an inclination angle of 40°, whereas the working fluid of methanol and the title angle of 30° were considered during the spring season. Based on the trial results, as shown in Fig. 14b and **c,** it is crystal clear that the PV/T system had lower rear panel temperature and generated more electricity. The average growth percentages of 7.7% and 5.67% in electrical power were obtained in summer and spring conditions, when compared to a single PV system. By contrast, photothermal efficiencies were 16.35% and 45.14%, respectively. The performance difference between the loop TPCT equipped with concentric double-tube condenser and non-concentric multi-tube condenser was compared by Yao et al. [86]. The results showed that the non-concentric type system performed better at higher latitude in northern China, with an increase of 17.76% in thermal efficiency. Habeeb et al. [87,88] performed field examinations to test a photovoltaic thermal module cooled using a TPCT. The copper plate was inlaid with the evaporation end of the TPCT to expand the heat absorption area, whereas the condensation end was cooled by a water jacket. Based on the results, the electrical efficiency of the cooled system could be enhanced by approximately 11–14% in comparison with the reference PV system.

Fig. 15 depicts a hybrid PV/T system combined with solar desalination technology reported in Ref. [89] to create fresh water. The author designed four different layouts (PV/T-TPCT-solar still (Fig. 15a), PV/T-TPCT-solar still using a fan (Fig. 15b), solar still (Fig. 15c), and PV/T-TPCT-solar still using a reflector (Fig. 15d)) to determine the optimal structure to produce pure water. In these systems, the surplus heat rejected by the solar cell is carried directly through the TPCT to heat the seawater. The total efficiency (thermal + electrical) for the PV/T-TPCT-solar still, solar still, and PV/T-TPCT-solar still using a reflector were 44.71%, 48.33%, and 44.64%, respectively. This innovative approach has been shown to be both successful and cost-efficient.

To address freezing issues encountered by traditional water-based PV/T systems under cold weather conditions, Ji and Pei's team conducted a series of systematic studies on heat pipe PV/T systems, from technical principles to practical operation, and they revealed this promising technology by tapping into its performance [90–92]. More specifically, they determined that a heat pipe charged with ammonia fluid can be employed in harsh winter regions, such as Beijing and Lhasa. Based on preliminary statistics, the authors reported that the conversion efficiency of the new system improved by 9.5%, 0.6% higher than the water thermosiphon system. Following the footsteps of Pei and Ji, Fu et al. [93] investigated a heat pipe PV/T system with varying pipe spaces and inclination angles. The spaces between two adjacent heat pipes were set to 8 cm and 14 cm, respectively, and the angles of each PV/T system were 32 ° and 45 °, respectively. A glass cover was placed above the PV panel to reduce the heat loss from the front side. Two water tanks with a volume of 200 L were used as storage tanks for the two independent

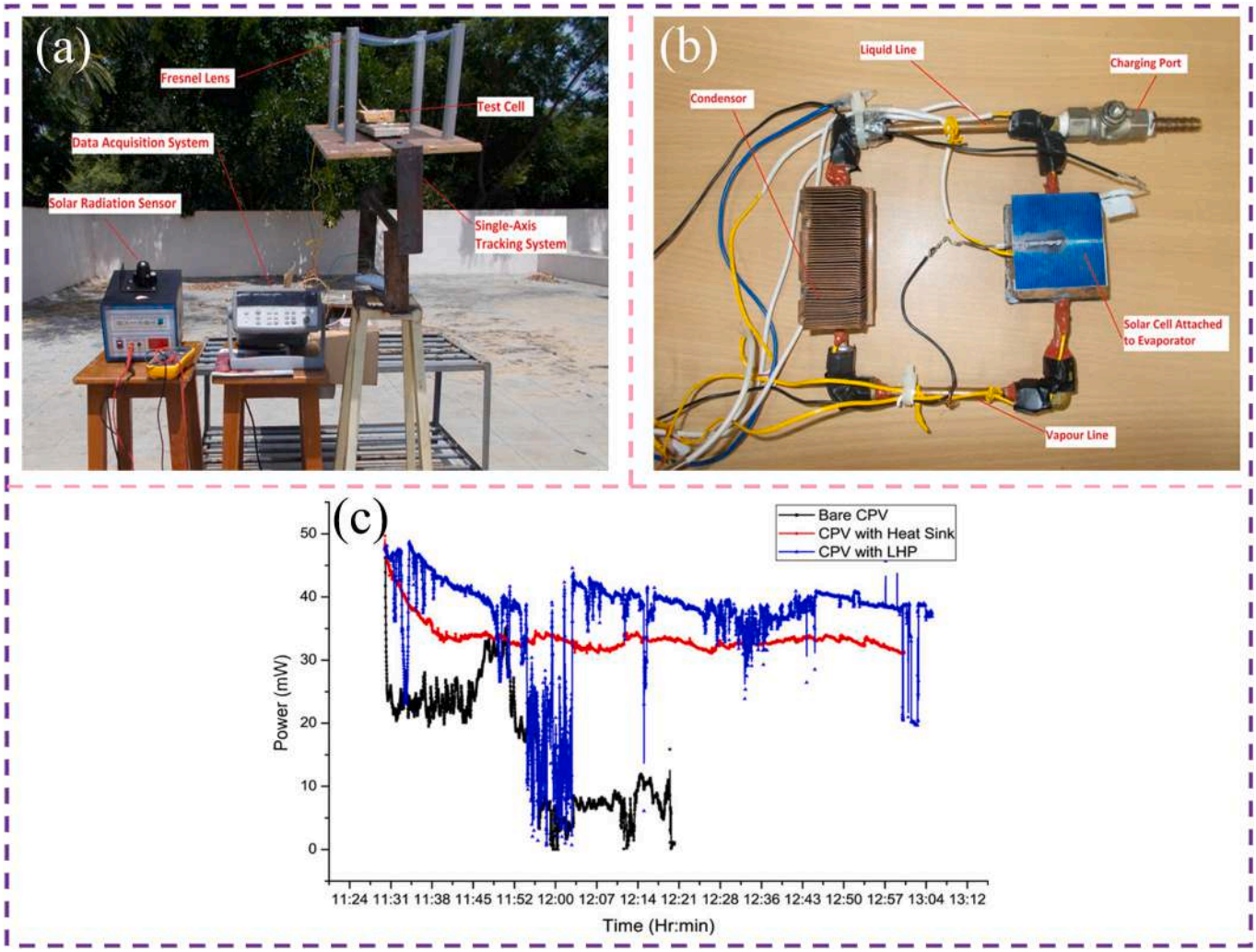



**Fig. 12.** Actual setup of (a) CPV system, (b) solar cell cooled by LHP, (c) power output under various conditions (adapted/reproduced with permission from Ref. [79]). Copyright © 2019 John Wiley and Sons.

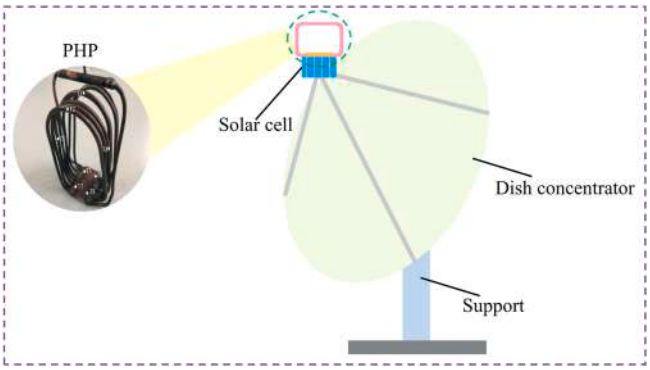



**Fig. 13.** Dish CPV system cooled by 3D PHP (adapted/reproduced with permission from Ref. [84]). Copyright © 2020 Elsevier.

systems. Based on these findings, a lower inclination angle and smaller pipe pitch results in better performance. Moreover, Pei et al. [94] discussed the effect of adding a glass cover to a heat-pipe PV/T system. Laminated solar cells with an effective total area of 0.728 $m^2$ were sealed within the aluminium box, and the space between the PV panel and the box was filled with insulation material, except for the front side. The systems, with and without a glass cover, were exposed to the same external environmental conditions. Their results showed that the performance variation of the glazed system was more stable than that of the unglazed system; however, the exergy efficiency of the glazed system was lower than that of the unglazed system.

MHPA equipment is favoured by many researchers because of its good structural adaptability and efficient heat-transfer performance. Hence, a significant portion of the heat pipe PV/T research is based on the MHPA. Tang et al. [95] conducted an comparative study between the air-cooled and water-cooled MHPA for cooling PV cell. In their test, an MHPA device with a length of 0.283 m and 0.3 m in width was attached to the PV panel. Consequently, the power output of the PV cell increased by 13.9% and 8.4% for water-cooled and air-cooled conditions, respectively. The study was an early attempt to combine a novel MHPA with a PV/T system to eliminate or minimize the large contact thermal resistance. The performance of the MHPA-PV/T system cooled by air flow, as shown in Fig. 16, was studied by Wang et al. [96]. The data collected showed that the MHPA cooling could achieve excellent temperature uniformity with a minimum temperature differential of 0.77 K and a maximum thermal and electrical efficiency of 65.90%.

Yuan et al. [97] also studied an MHPA-PV/T system under higher ambient temperature conditions. In their investigation, three distinct layouts (specifically, the MHPA-PV/T system, water-cooled PV/T system, and PV-only system) were evaluated simultaneously under blazing environmental conditions (Fig. 17a). The connections between the aluminium plate and heat exchangers were not the same. The collecting

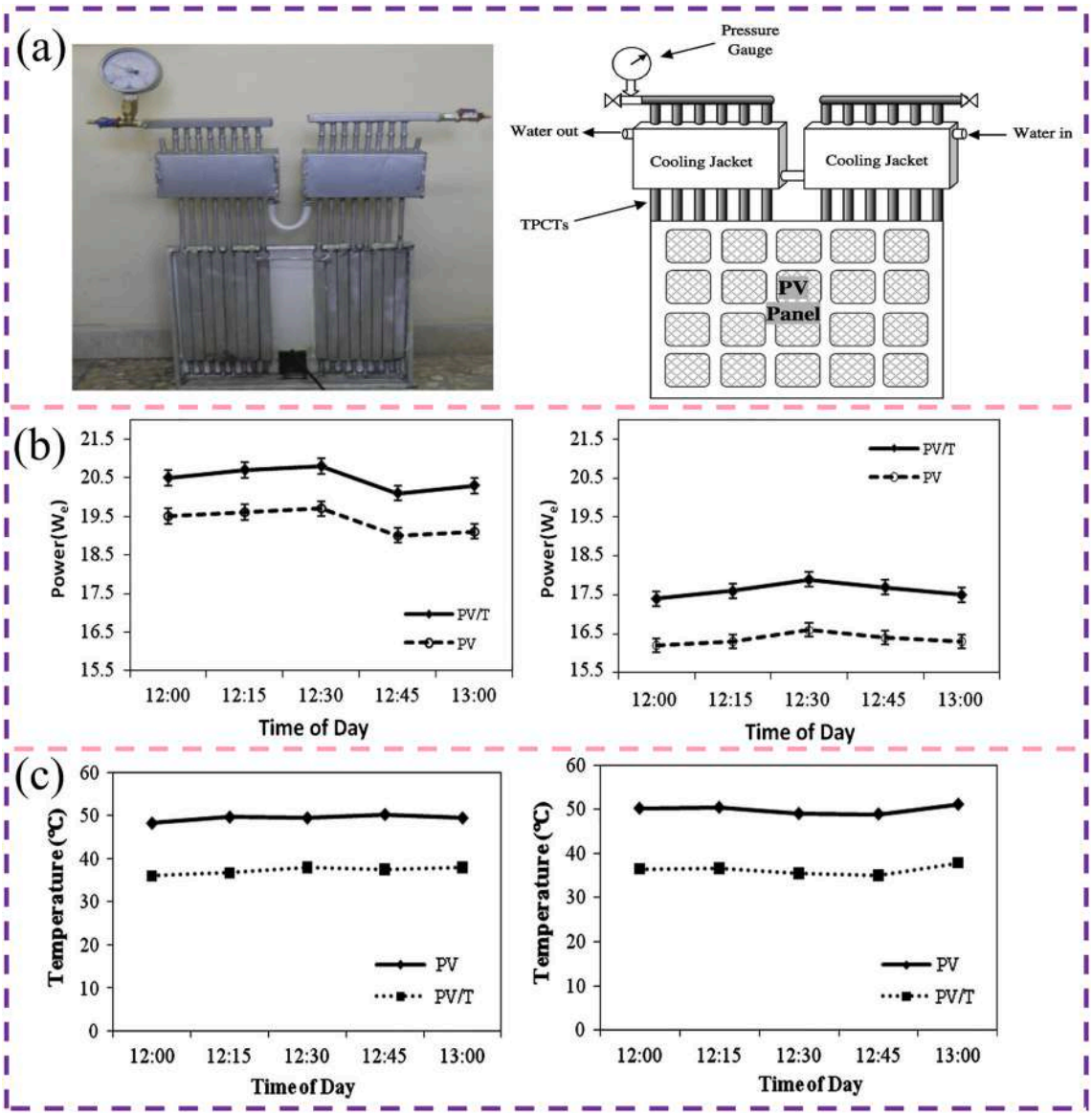



**Fig. 14.** Results from the TPCT-PV/T system (a) experimental setup, (b) power output in spring and summer, (c) variation of PV temperature in spring and summer (adapted/reproduced with permission from Ref. [85]). Copyright © 2014 Elsevier.

pipes of the water-cooled PV/T system were welded to a back plate, and the MHPA was bonded to the backplate using thermally conductive glue. Results revealed that the electrical efficiency of the reference PV panel was found to be approximately 8.6%–7.0%, and 11.2%–10%, 9.6%–7.7% for the water-cooled module and MHPA cooled. Chen et al. [98] and Kai et al. [99] conducted a series of experimental investigations on a low-concentration (LCPV/T (LCPV/T) system integrated with MHPA (Fig. 17b). In addition, an optical compound parabolic concentrator (CPC) with a two-axis tracking mechanism was placed at the margins of the solar cells to reflect the incoming sunbeam to the receiver, resulting in a concentration ratio of up to 4 × . The average electrical and thermal efficiencies of the developed system are 10% and 50%, respectively, and the maximum exergy efficiency of the generated electricity was 14.49%.

Lee et al. [100] designed a cooling chamber combined with a heat pipe to minimize the operating temperature of solar cell arrays. Sixteen triple-junction solar cells were mounted on a 4 × 4 matrix, each of which was installed with a Fresnel lens. A thermal absorber integrated with a heat pipe was placed underneath the solar cell to extract the waste heat. Running water was utilised to remove the unnecessary heat in the cooling chamber to produce hot water, as depicted in Fig. 17c. According to their findings, the tested PV/T system could attain an overall efficiency of 97%.

Several key parameters that may affect the operation of the heat-pipe PV/T system have also been considered by researchers. Zhang et al. [101] analyzed a heat-pipe PV/T system furniture with different tank volumes. A simulation model was also prepared by the authors using TRNSYS software to optimise the layout design. The results revealed that the system reached the highest efficiency of 67.5% when the water tank volume was 80 L. Moreover, Wang et al. [102] performed a laboratory test to assess the impact of employing wickless and wire-meshed heat pipes in PV/T systems. The primary findings revealed that the wickless heat pipe system was more sensitive to the tilt angle than the wire-meshed heat pipe system. A comprehensive technology combined with a high-concentrating photovoltaic/thermal (HCPV/T) and air-conditioning system was proposed by Hu et al. [103], as shown in Fig. 17d. The coupled system achieved a total efficiency and energy savings of 49.77% and 15.40%, respectively. The filling ratio is a critical parameter that could influence the operational performance of a heat-pipe-based PV/T system. Zhang et al. [104] carried out plenty of tests with different heat pipe filling ratios and they found that the ideal

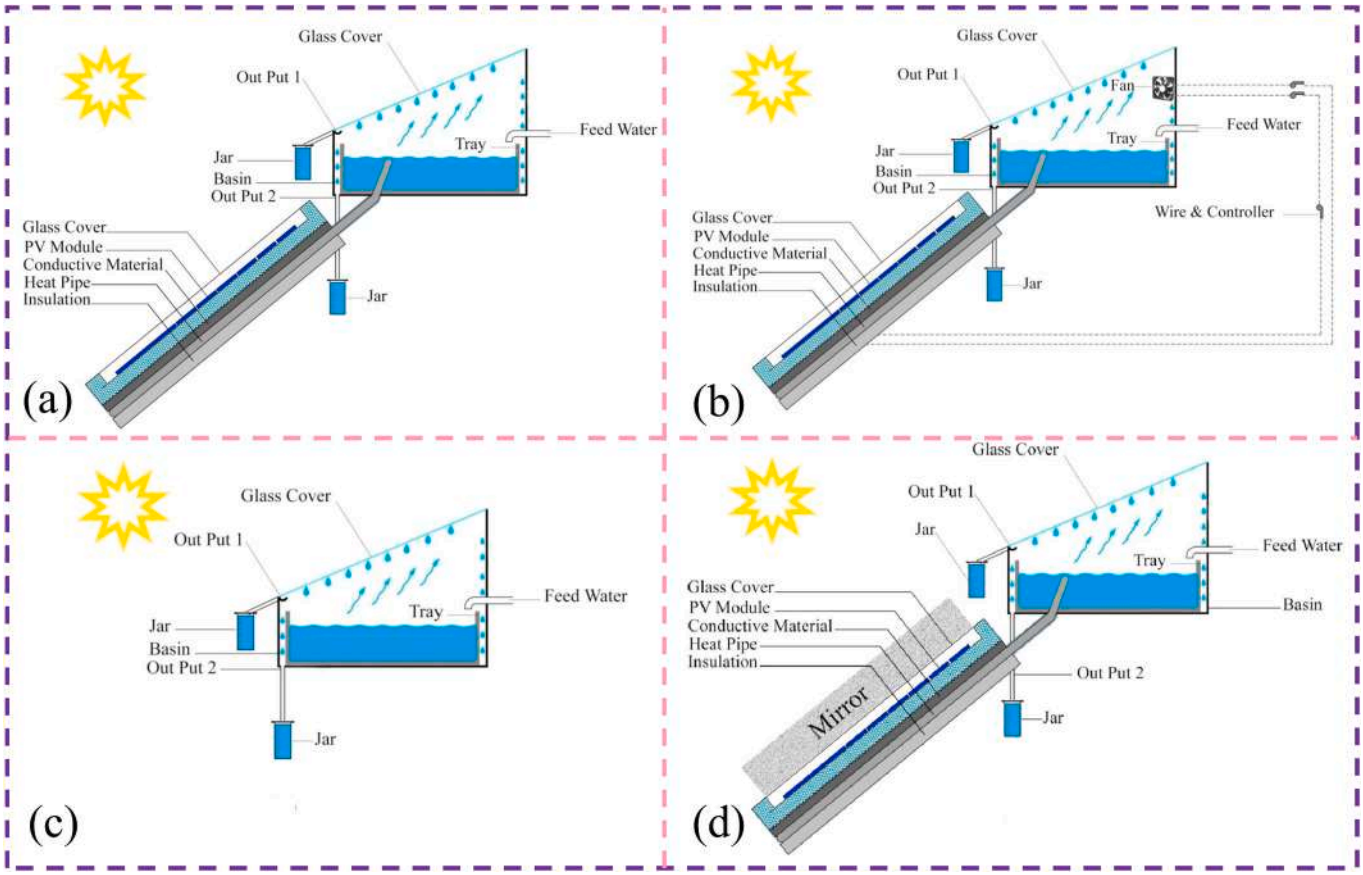



**Fig. 15.** Coupled system used for desalination, (a) PV/T-TPCT-solar still, (b) PV/T-TPCT-solar still using fan, (c) solar still, and (d) PV/T-TPCT-solar still using reflector. (adapted/reproduced with permission from Ref. [89]). Copyright © 2018 Elsevier.

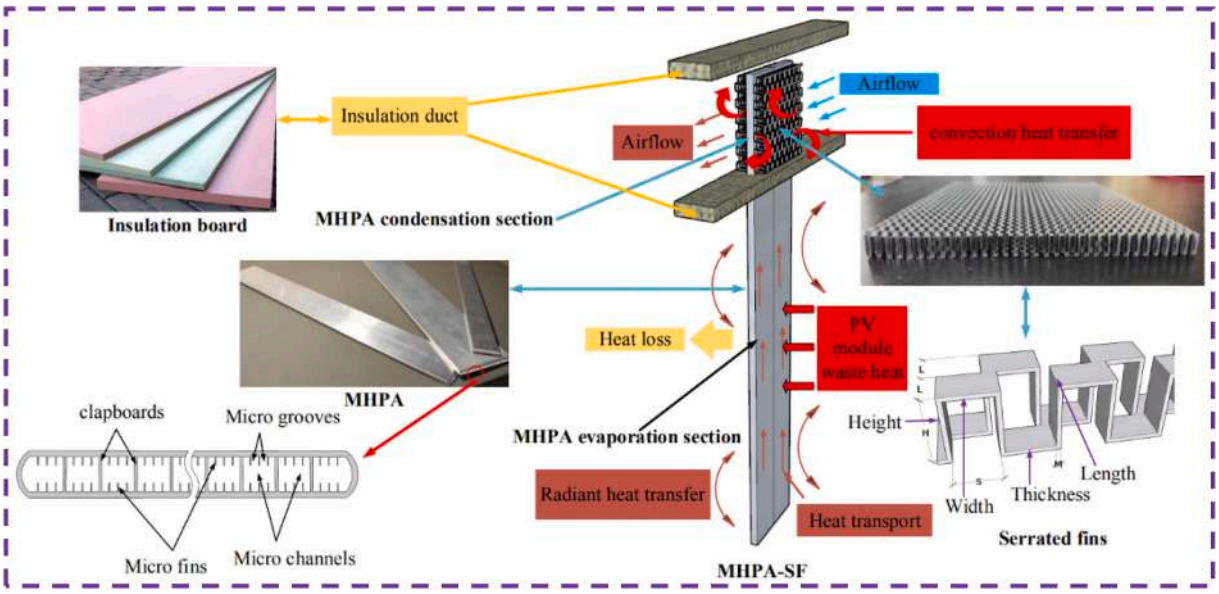



**Fig. 16.** The schematic of the air-cooled MHPA-PV/T system. (adapted/reproduced with permission from Ref. [96]). Copyright © 2022 Elsevier.

filling range was approximately 32–40%. In addition, Zhang et al. [105] established a numerical model to describe the effect of the tilt angle on a heat pipe-based PV/T system. The deviation between the developed model and the experimental data was modest and could be used to describe the performance of such a system.

Table 1 summarises the various representative collector structures of the heat pipe PV/T. A vertical comparison with other articles is required to more effectively showcase the features of diverse articles. To establish a vertical comparison, Table 2 presents a few example publications from the year, study methods, PV types, concentrating technology, main conclusions, and highlights.

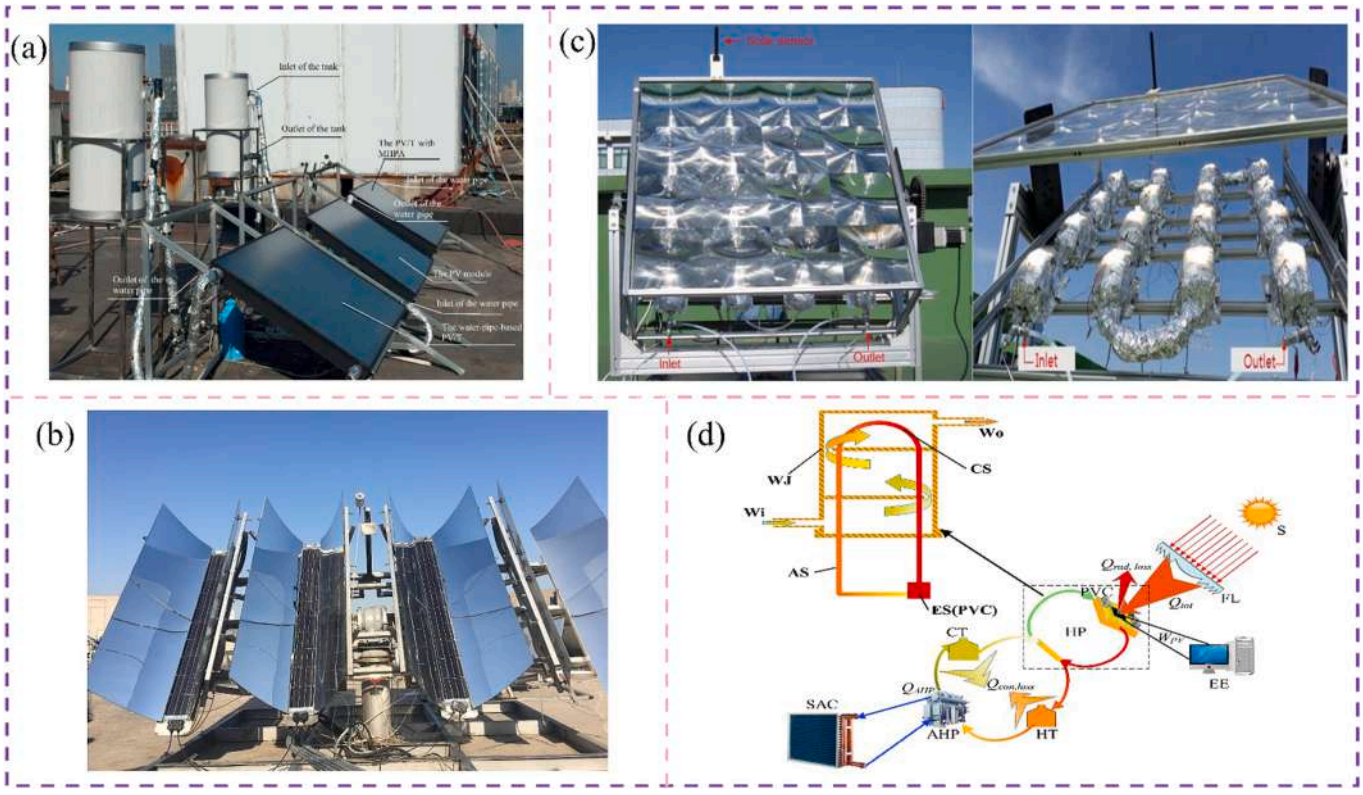



**Fig. 17.** Typical heat pipe PV/T systems (a) actual picture of the MHPA-PV/T system (adapted/reproduced with permission from Ref. [97]). Copyright © 2018 Elsevier. (b) LCPV/T system combined with MHPA (adapted/reproduced with permission from Ref. [98]). Copyright © 2018 Elsevier. (c) array type heat pipe PV/T system (adapted/reproduced with permission from Ref. [100]). Copyright © 2015 John Wiley and Sons. (d) the hybrid HCPV/T system combined with air conditioning technology (adapted/reproduced with permission from Ref. [103]). Copyright © 2019 Elsevier.

## 4. Development of heat pipe PV&PV/T systems combined with innovative technologies

Recently, scientists have dedicated their efforts to introducing novel technologies to combine with heat pipe PV&PV and PV/T systems. Accordingly, academics have consistently acknowledged the concept of hybrid technologies integrating multiple technological elements, who have also been at the forefront of the recent innovative revolution. Prior to analysing future analytical work, it is necessary to present a succinct description of the most recent technologies and emphasise their unique characteristics. This chapter defines hybrid heat pipe PV&PV/T systems integrated with TE technology, PCM technology, heat pump technology, BIPV technology, and nanotechnology, as illustrated in Fig. 18.

### 4.1. Combined with TE technology

The TE device is a type of bidirectional energy conversion unit that could function in two modes: electricity-generating mode or active refrigeration mode [118]. The solid-state features of a TE units offer several noticeable benefits, including no moving parts, a superior modular design concept, no environmental harm, low maintenance costs, long life cycle, excellent scalability, and noise-free operation [119–121].

The Seebeck effect is the theoretical basis for TE generators converting thermal energy into electrical energy. This conversion process, discovered by Thomas Seebeck in 1821, may be characterised as follows: current and electrostatic forces can be detected in a specific circuit if a temperature gradient ($\Delta T$) occurs between the junction of two distinct metals. The relationship between the temperature gradient and voltage difference is defined as the Seebeck coefficient ($\alpha$) using the following equation:

$$\alpha = \Delta V / \Delta T \tag{1}$$

where, $\alpha$, $\Delta V$, and $\Delta T$ denote the Seebeck coefficient, voltage difference, and temperature gradient, respectively.

Peltier effect, the fundamental of TE cooler discovered by Jean-Charles Peltier in 1834, is another type of thermoelectric effect. The difference between the Fermi energies of different conductor materials is the essence of this phenomenon [120]. Generally, the temperature of the junction and the intrinsic properties of different conductor materials govern the quantity of heat released or absorbed during this process. Moreover, the figure of merit (ZT), a dimensionless parameter derived from Eq. (2), is used to characterise the performance of thermoelectric materials [122].

$$ZT = \alpha^2 \sigma T / \kappa \tag{2}$$

where $\alpha$, $\sigma$, $\kappa$, and $T$ denote the Seebeck coefficient, electrical conductivity of the materials, thermal conductivity of the materials, and temperature, respectively. The TEG efficiency can be represented as follows:

$$\eta_{\max} = \frac{T_H - T_C}{T_H} \cdot \frac{\sqrt{1+ZT}-1}{\sqrt{1+ZT}+\frac{T_C}{T_H}} \tag{3}$$

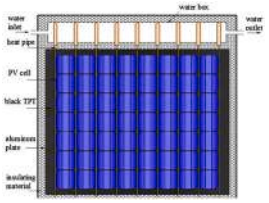
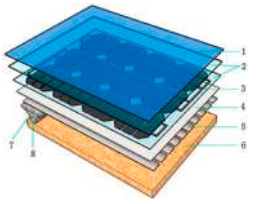
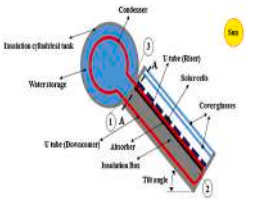
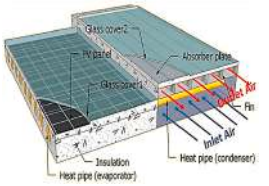
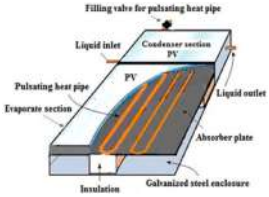
In this equation, the first term of the equation, $\frac{T_H - T_C}{T_H}$, is called Carnot efficiency. $T_H$ and $T_C$ indicate the temperatures of the hot and cold ends, respectively.

The maximum coefficient of performance (COP) of the TE cooler is calculated using the following equation:

$$COP_{\max} = \frac{T_C}{T_H - T_C} \cdot \frac{\sqrt{1+ZT}-\frac{T_H}{T_C}}{\sqrt{1+ZT}+1} \tag{4}$$

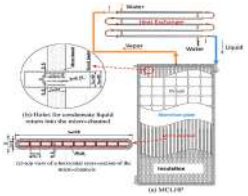
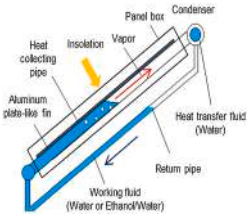
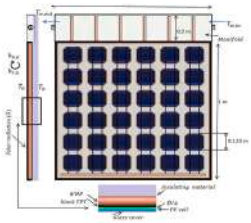
The combination of TE and PV technologies is a win-win situation,

**Table 1**
Summary of some representative structure of the heat pipe PV/T collectors.

| References | Descriptions | Schematic view of the system | Method | Types of heat pipe | Working fluids | Major observations |
|---|---|---|---|---|---|---|
| [24,106,107] | ● A novel type of CTHP based PV/T system was designed and tested from numerically and experimentally aspect. |  | Experimental & simulation | CTHP | Water: for ice-free condition | ● The daily thermal, electrical efficiencies, heat gain and electrical gain of the system were 41.9% and 9.4%, 276.9 W/m$^2$ and 62.3 W/m$^2$, respectively. |
| | ● One of the distinct features of this kind of PV/T system is the anti-freezing characteristic. | (adapted/reproduced with permission from Ref. [92]). Copyright © 2012 Elsevier. | | | R134a: for cold condition | ● The proposed PV/T system can operate stably in severe cold areas without considering the freezing issues. |
| [108,109] | ● MHPA device was integrated with conventional PV/T systems. |  | Experimental & simulation | MHPA | Acetone | ● This structure can provide less thermal contact resistance compared with other forms of heat pipe PV/T collector, and it is easy to massive production. |
| | ● The MHPA device was closely and directly contacted with the PV module. | (adapted/reproduced with permission from Ref. [109]). Copyright © 2016 Elsevier. | | | | ● The remaining features can be summarized as: better temperature uniformity, higher heat transfer performance, more reliable, lower pressure drop (compared to conventional PV/T system). |
| [110] | ● Loop mini TPCT was introduced in PV/T system. |  | Experimental & simulation | TPCT | Water | ● The temperature of water in the storage tank can reach up to 345.15 K at night. |
| | ● Integrated design, the water tank is placed directly above the PV module. | (adapted/reproduced with permission from Ref. [110]). Copyright © 2016 Elsevier. | | | | ● The optimal distance between the glass cover and solar cell was 10 mm. |
| [111] | ● A dual channels solar air collector was incorporated with conventional PV/T system by using heat pipe. |  | Experimental & simulation | CTHP | Water | ● The proposed system had a maximum thermal efficiency of 69.2%. |
| | ● A complete economic analysis was carried out. | (adapted/reproduced with permission from Ref. [111]). Copyright © 2019 Elsevier. | | | | ● The payback time is between 5.7 and 16.8 years<br>● The temperature difference of the heat pipe cooled system was 11.6 K lower than the reference system, which shows a good temperature distribution. |
| [112] | ● PHP apparatus was proposed to combine with PV/T system. |  | Experimental | PHP | Distilled water | ● The best performance of the hybrid system happened at noon with the filling ratio and inlet velocity of 45% and 0.05 kg/s. |
| | ● The influence of various parameters (including solar radiation, inlet temperature, title angle, filling ratio, ambient conditions) were detailly discussed. | (adapted/reproduced with permission from Ref. [112]). Copyright © 2018 Springer Nature. | | | | ● The developed system showed better performance in comparison with conventional PV/T and PV systems. |

(*continued on next page*)

**Table 1** (*continued*)

| References | Descriptions | Schematic view of the system | Method | Types of heat pipe | Working fluids | Major observations |
|---|---|---|---|---|---|---|
| [113–115] | ● Micro channel LHP with coaxial tube heat exchanger was employed in PV/T system.<br>● The impact of inlet temperature, ambient temperature, coolant flow rate was analyzed. |  <br>(adapted/reproduced with permission from Ref. [113]). Copyright © 2019 Elsevier. | Experimental & simulation | LHP | R134a | ● The variation ranges of the electrical efficiency and thermal efficiency were 15.59–18.34% and 25.5–62.2%, respectively.<br>● The enhancement of the total energy conversion efficiencies was 17.20% and 33.31% higher, compared with the PV/T system and BIPV/T system. |
| [116] | ● An eco-friendly PV/T system integrated with decompression boiling heat collector was proposed. |  | Experimental | TPCT | Ethanol/ water solution (10, 20 and 30 wt %) | ● Even in outdoor conditions in summer, the temperature of PV panel could be kept about 45 °C.<br>● The outlet temperature could be higher than 60 °C by adjusting the flow rate of the coolant. |
| [45] | ● A wickless heat pipe filled with acetone was adapted in PV/T system.<br>● The real meteorological condition was taken into account in numerical study. |  <br>(adapted/reproduced with permission from Ref. [45]). Copyright © 2021 Elsevier. | Experimental & simulation | TPCT | Acetone | ● The average power generation was 25.28 W and 72.49 W in winter and summer conditions.<br>● The values of thermal power were 69.3 W and 77.19 W, respectively, under the same conditions.<br>● The performance of the system varied with the number of heat pipes, and the optimal value is 10 when the system shows the best results. |

gaining traction. The waste heat accumulated on the backside of the PV panel can be used as a heat source for the TE generator to generate electrical energy. From the perspective of the entire system, this synergy has had a positive influence on two fronts.

- First, the heat of the PV panel can be dispersed over time and the temperature can be maintained within an appropriate range, resulting in greater electric energy production.
- Second, this portion of the increased heat may be used to power the TE generator, thereby considerably increasing the total power generating capacity and improving the overall power generation capacity of the system.

Furthermore, if the TE device is in active cooling mode, the electricity generated by the PV module can be utilised directly to power the TE cooler owing to the direct current electric source [123]. The feasibility of combining TE with PV technologies has been thoroughly demonstrated and explored, the most popular of which are direct-coupling integration and spectrum-splitting techniques [124–127].

Introducing heat pipe technology with a PV-TEG system is a neophilia that has emerged in recent years, and the majority of works currently focus on establishing a mathematical model or conducting lab-scale investigations.

Li et al. [128] developed a mathematical model to characterise the performance of an MHPA-based PV-TEG system under various climatic conditions, and the proposed system was compared with a standalone PV arrangement, as shown in Fig. 19. A solar cell with a receiving area of 0.216 $m^2$ was located on the evaporator of the MHPA heat exchanger, and the hot side of the TEG unit was connected to the condenser of the MHPA. The MHPA setup acted as a "thermal bridge" to transport useless heat from the PV to the TEG device. The results demonstrated that the hybrid PV-TEG system outperformed the PV system in terms of generated power output and economic performance. Based on prior numerical analysis, the author developed an experimental work under real-world settings in Refs. [32,129] (shown in Fig. 20a). The experimental results revealed that a 14% increase in electrical efficiency was attained compared to a pure PV system, and the influence of various environmental conditions was also investigated. As illustrated in Fig. 20b and **c**, the hybrid system exhibited a higher power output under a wind speed of 7 m/s compared to the case of 1 m/s. This preliminary study provides guidance for the design of similar systems and experimentally demonstrates the feasibility of the system.

Shittu et al. [130] also conducted a validation experiment to establish the viability of a hybrid PV-TEG-MHPA system, as shown in Fig. 21. An active water-cooled block was utilised as the heat spreader instead of the passive finned heat sink to enhance the heat transfer effect and collect hot water. It can be concluded that the newly proposed arrangement is both practicable and effective, primarily because the obtained highest thermal efficiency can reach up to 69.53%. In another

**Table 2**
Summary of the heat pipe PV& PV/T systems.

| Author(s) | Year | Methods | Heat pipe types | PV types | Concentrated technology | Results | Highlights/Findings |
|---|---|---|---|---|---|---|---|
| Akbarzadeh et al. [65] | 1996 | Experimental | TPCT | polycrystalline silicon | Parabolic reflector | The maximum power output was 20.6 W. | ● A 50% PV cell temperature drop was observed by using the heat pipe cooling method. |
| Anderson et al. [69] | 2008 | Experimental & numerical | CTHP | / | / | The system could deal with the incoming heat flux of 40 W/m$^2$. | ● A combined experimental and numerical method was conducted to study the CPV cell cooled by finned-heat pipe under natural convection. |
| Kuo et al. [68] | 2009 | Experimental | CTHP | InGaP/GaAs/Ge | Fresnel Lens & secondary concentrator | The generated power was 30 W/cm$^2$ under passive thermal management. | ● A novel designed solar engine setup was tested under high concentration ratio conditions. |
| Pei et al. [91] | 2010 | Experimental | TPCT | Monocrystalline silicon | / | The electrical efficiency of heat pipe PV/T system was 6.7% higher than the water type layout. | ● A comparative test between the heat pipe PV/T system and water type PV/T system was performed. |
| Huang et al. [72] | 2012 | Experimental | MHPA | InGaP/GaAs/Ge | Fresnel Lens | Energy conversion efficiency improved by 3.1%. | ● The results of using heat pipes in CPV system with various inner wick structures and working fluids were studied and summarized. |
| Gang et al. [92] | 2012 | Experimental & numerical | CTHP | Monocrystalline silicon | / | The maximum accumulated thermal energy was 1872.22, 3328.25 and 2352.95 MJ/m$^2$ in typical regions. | ● The proposed system can be applied in the cold area without considering the icing problems. |
| Geng et al. [82] | 2012 | Experimental & numerical | PHP | / | Parabolic reflector | Using the PHP heat sink is feasible to cool down the solar cell. | ● The experimental results indicated that the cooling system could run well, no more than 50X suns. |
| Wang et al. [73] | 2013 | Experimental | TPCT | InGaP/GaAs/Ge | Dish concentrator & secondary concentrator | 24.76% in electrical efficiency: two-stage<br>21.12% in electrical efficiency: one-stage | ● The performance of the system was dominated by the factors of solar radiation and the temperature of the solar cell. |
| Armijo et al. [76] | 2013 | Experimental | TPCT | InGaP/GaAs/Ge &monocrystalline silicon | / | Electrical efficiency enhanced by 1.7% for PV cells.<br>Electrical efficiency enhanced by 2.1% for CPV cells. | ● The employment of heat pipe in cooling conventional solar cell and the concentrated solar cell was feasible.<br>● The conventional solar cell and concentrated solar cell could operate under 132X and 560X suns with the assistance of the heat pipe. |
| Wang et al. [117] | 2015 | Numerical | MHPA | Monocrystalline silicon | / | The electrical, thermal and overall efficiencies of the system in different areas were obtained. | ● A numerical model was developed to predict the performance of the MHPA-PV/T system worked in various cities in China. |
| Deng et al. [108] | 2015 | Experimental | MHPA | Monocrystalline silicon | / | The annual average electrical efficiency of the MHPA-PV/T configuration was 13.51%, and the related thermal efficiency was 26.80%. | ● A whole year experimental test was conducted to study the novel type of MHPA-PV/T system. |
| Chen et al. [74] | 2016 | Experimental & numerical | TPCT | / | / | The cooling system charged with acetone showed the best performance. | ● The use of various working fluids in TPCT was experimentally and numerically studied. |
| Du et al. [58] | 2017 | Experimental | MHPA | Amorphous silicon | / | A significant temperature reduction of 47–50% was observed. | ● The overall performance of the system can be furtherly improved by integrating the water box with the heat pipe. |
| Modjinou et al. [46] | 2017 | Experimental & numerical | MHPA | Crystalline silicon | / | The produced electricity was found to be 70 W, with an average thermal efficiency of 50.7% and electrical efficiency of 7.6%. | ● A transient numerical model was developed to describe the behavior of the MHPA-PV/T system, and the model agreed well with the tested data. |
| Alizadeh et al. [61] | 2018 | Numerical | PHP | Monocrystalline silicon | / | Using PHP setup led to a 16.1 K temperature reduction for PV cell, 18% improvement of power output was also identified. | ● The optimal filling ratio of 40% was confirmed as well as the inclination angle. |
| Chen et al. [98] | 2018 | Experimental | MHPA | Crystalline silicon | Parabolic reflector | The photothermal and photoelectric efficiencies were 54.48%, 14.49%, respectively. | ● A practical and comprehensive examination of the MHPA-LCPV/T system was conducted. |
| Anand et al. [79] | 2019 | Experimental | LHP | Polycrystalline silicon | Fresnel Lens | LHP cooling showed the best results, with an average cell temperature of 310.95 K. | ● 130% and 10% enhancement in power output were obtained compared with the water-cooled and no-cooled conditions. |

(*continued on next page*)

**Table 2** (*continued*)

| Author(s) | Year | Methods | Heat pipe types | PV types | Concentrated technology | Results | Highlights/Findings |
|---|---|---|---|---|---|---|---|
| Soliman et al. [80] | 2019 | Experimental | MHPA | Polycrystalline silicon | / | The cooling efficiency of 36.3% was gained, while the increase of electrical efficiency was 13.6%. | ● Under the solar intensity of 500W/$m^2$, the electrical generation could be enhanced by 24.3% when the non-cooled condition was compared. |
| Zhang et al. [104] | 2019 | Experimental | TPCT | Monocrystalline silicon | / | The optimized result of the filling ratio was about 32%. | ● A detailed investigation on the impact of various filling ratios (ranging from 24% to 48%) on the performance of the HP-PV/T system was studied. |
| Wang et al. [84] | 2020 | Experimental | PHP | / | Dish concentrator | Under the 58.8X suns, the temperature of the solar cell could be maintained no more than 330.15 K. | ● A 3D PHP used for cooling solar cells with various filling ratios and different inner structures was investigated. |
| Al-Amri F et al. [18] | 2021 | Experimental | / | Polycrystalline silicon | / | The best cooling efficiency could reach up to 53%. | ● The condenser part of the heat pipe immersed in various materials (PCM, oil, water) was tested. |
| Ren et al. [114] | 2021 | Experimental & numerical | MHPA | Amorphous silicon | / | When the thickness of the air gap was 15 mm, the overall conversion efficiency was 47.61%. | ● The influence of the thickness of the air gap between the glass cover and the PV/T panel was discussed. |

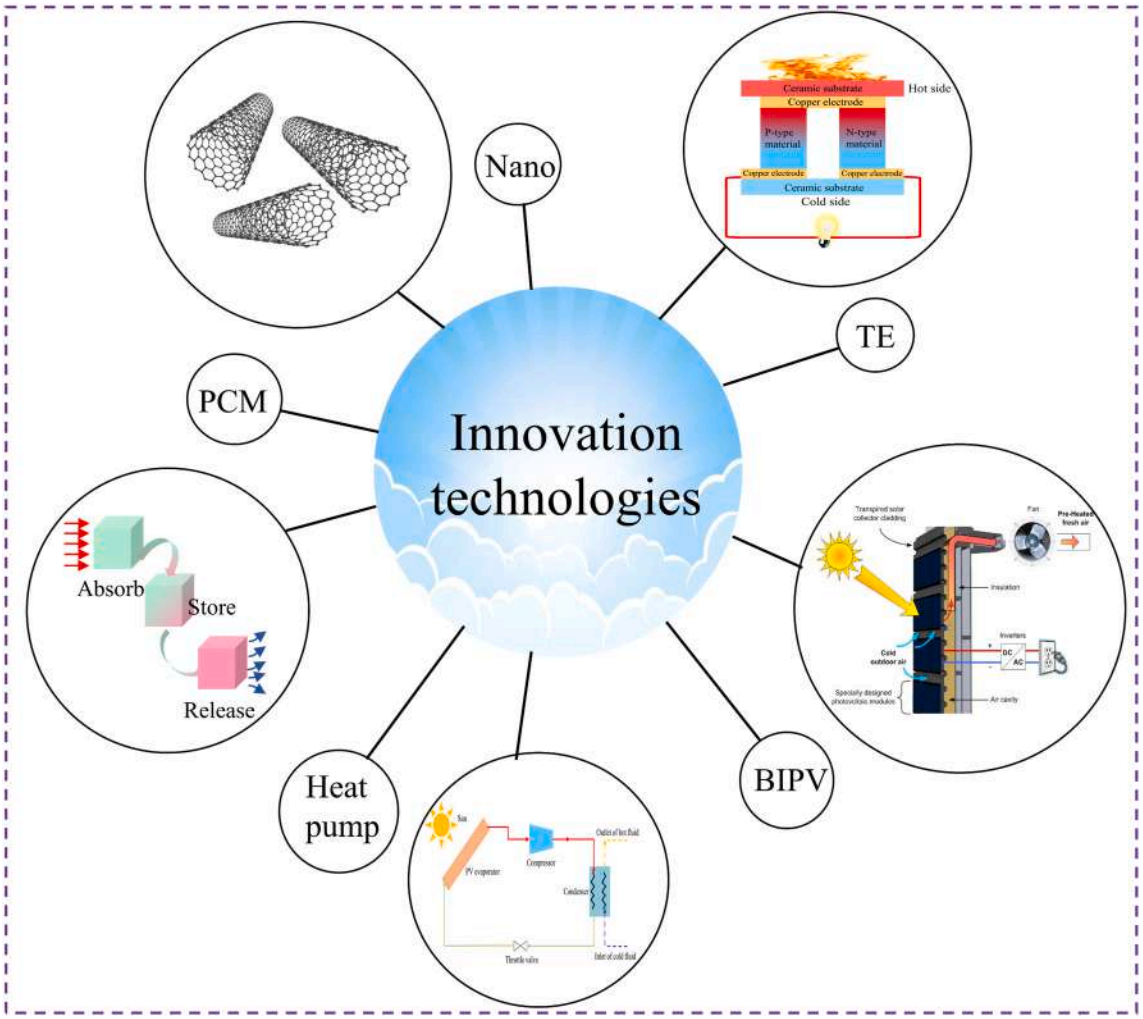



**Fig. 18.** Combined technologies (a) TE technology, (b) BIPV technology, (c) PCM technology, (d) heat pump technology, (e) nanotechnology.

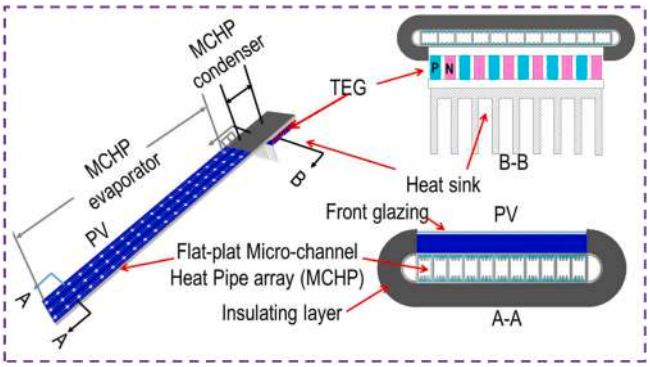



**Fig. 19.** Schematic view of the PV-TEG system combined with MHPA (adapted/reproduced with permission from Ref. [128]). Copyright © 2016 Elsevier.

study, Shittu et al. [131] conducted a comparative study of a CPV-TEG system and a CPV system with a heat pipe as a passive thermal management configuration. The results indicated that under 6 × concentrations, the performance of the heat pipe cooled CPV-TEG and CPV systems were enhanced by 1.47% and 61.01%, respectively.

Makki et al. [133] conducted a theoretical study that compared the performance of a PV-TEG system to that of a normal PV system using a CTHP as a heat sink. Based on numerical analysis, the authors concluded that using a TEG module can increase the total output of the system. In a normal PV-TEG system, the TEG unit is typically suspended at the bottom of the PV cell; however, the poor thermal conduction of the TEG materials hinders the heat conduction from the PV cell to the target heat sink. To overcome this problem, Gao et al. [132] developed a novel bifacial PV-STEG heat pipe system to address the mismatch problem of a typical tandem PV-TEG system, as illustrated in Fig. 22a. In this layout, the heat pipe was inserted between the PV unit and TEG module, and the special design allowed the heat flux from both sides of the evaporator. Two optical concentrators were utilised to provide concentrated light for the PV cell and STEG unit. From Fig. 22b-**22d**, it is evident that the proposed system exhibits the best results considering the generated electricity, efficiency, and average temperature. When compared to the tandem system, the newly proposed system had better power output and electrical efficiency, with specific values of 20.98% and 14.05%,

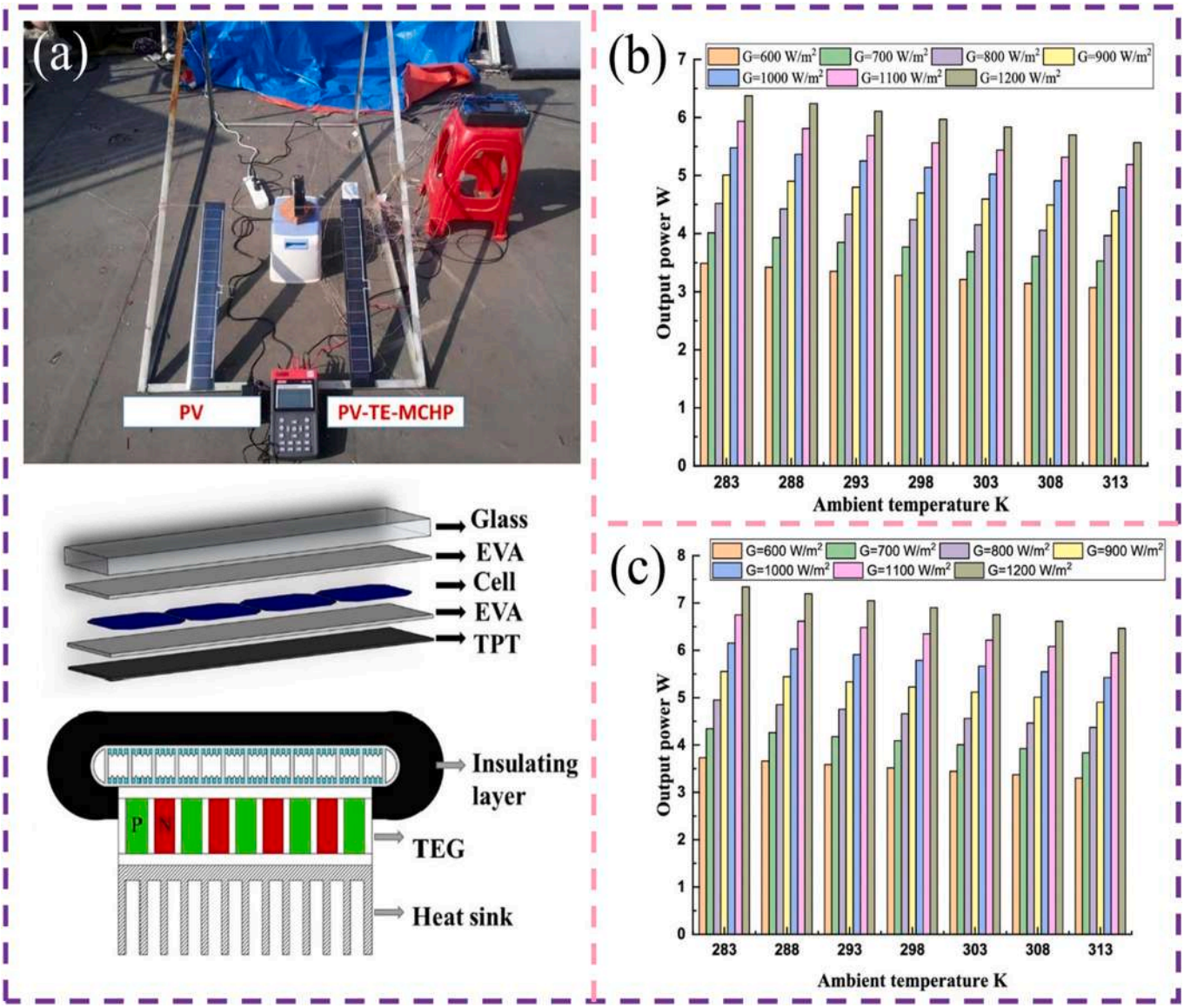



**Fig. 20.** PV-TEG system combined with MHPA (a) Actual picture of the system and detailed structure, (b) Output power under different ambient conditions (1 m/s and 7 m/s, from top to bottom) (adapted/reproduced with permission from Ref. [129]). Copyright © 2019 Elsevier.

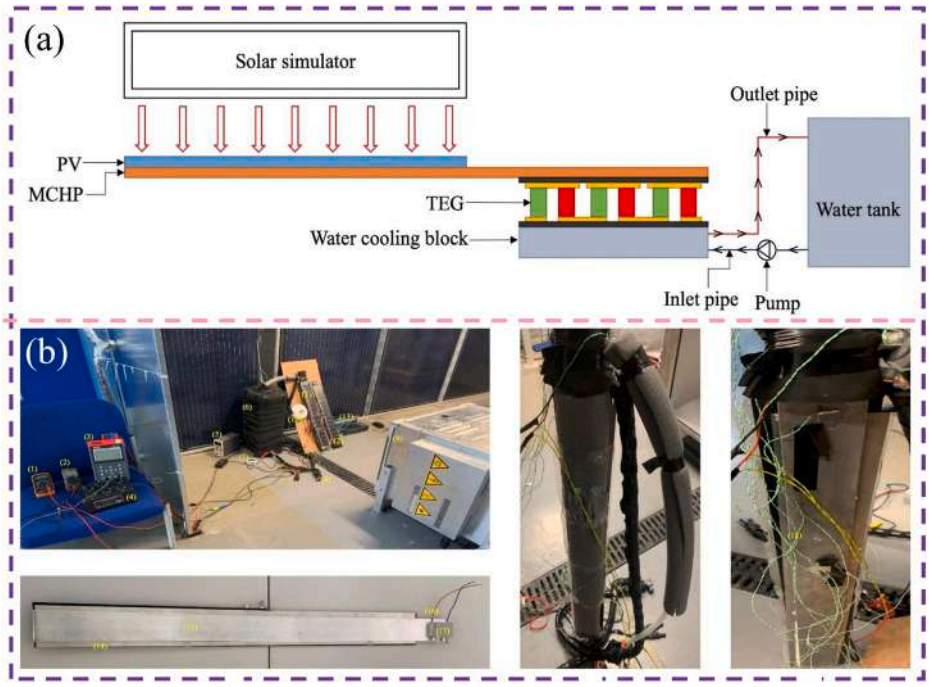



**Fig. 21.** Indoor test of the PV-TEG-MHPA system (adapted/reproduced with permission from Ref. [130]). Copyright © 2020 Elsevier.

respectively.

He et al. [134] investigated a new heat-pipe PV/T system with TE cooling and heating. The operational performance was evaluated under both winter and summer conditions, with average thermal and electrical efficiencies of 23.5% and 16.7%, respectively. The COP of the TE device was determined to be 1.7. Chen et al. [135] explored an LCPV/T system coupled with TEG and MHPA devices. The outdoor tests indicated that the daily average electrical efficiencies of the PV module and TEG setup were 11.8% and 0.23%, respectively, and overall efficiency of 57.03% was also achieved (Fig. 23a). The numerical model used to describe the entire process agreed well with the experimental results, as shown in Fig. 23b.

Fathabadi et al. [136] proposed a co-power generation system using a PV panel and TEG device. In this configuration, the role of the heat pipe was to transfer the collected solar energy to the TEG device, and the system was also designed to be easily combined with other power-generating forms (e.g., fuel cell or wind power generating devices). Wen et al. [137] compared two linear Fresnel PV/T-TEG systems with and without a TPCT. The numerical results revealed that the overall efficiency and electrical performance of the TEG could be effectively improved by the TPCT. In another study [138], as shown in Fig. 24a and **b**, the authors discussed the effect of adding aerogel glazing to the LFPV/T-TEG system. The influence of the thickness of aerogel materials on electrical energy and efficiency is presented in Fig. 24c and **d**. A downward and upward trend of PV electrical efficiency was observed when the thickness of aerogel increased from 10 mm to 70 mm. In addition, the measured TEG power indicated that employing aerogel glazing could increase the power output by 6 kJ.

Currently, the low energy conversion efficiency of the TE module is the major challenge to this system. The temperature matching characteristics between different solar cells and thermoelectric devices need to be further defined, and new system structures also need to be developed to address the current operating temperature issues. In addition, although the generated electricity is small for a single TEG module, the power could be amplified by connecting several TEG modules in parallel or in series. However, the overall cost of the TEG modules is still too high.

### 4.2. Combined with BIPV/T technology

The concept of building integrated photovoltaic (BIPV) technology, from the technical design of a green building or low-carbon building, is a combination method that uses PV units to replace conventional building materials or envelopes, such as building roofs, façades, daylighting, and windows [139]. When a PV module is incorporated into a structure, the initial costs of building materials and labour can be reduced compared with a standard non-integrated building system [140]. Moreover, the additional feature of power generation can eliminate or reduce the consumption of fossil fuels for artificial lighting, ventilation, heating, hot water supply, and air conditioning, leading to an eco-friendly environmental impact throughout the entire lifecycle and maintenance of building operation. Based on PV progress, the PV/T systems that have arisen can not only create energy but also collect the waste heat. When this layout is used on a building, it is referred to as a building integrated photovoltaic/thermal (BIPV/T) system. Because of the more complete usage of solar energy, the dual functions of the BIPV/T system in acquiring thermal and electricity results in higher performance to the BIPV system.

Urbanisation and population growth are accelerating. Additionally, half of the global population, or approximately 55%, reside in cities, and this number is expected to rise by a further 13% by 2050 [141]. Historically, building energy use has accounted for 30% of total global energy consumption and a third of associated $CO_2$ emissions, and this ratio is expected to further increase in the future. Low- or zero-energy buildings should play a larger role in resolving the tension between expanding urbanisation and the increased demand for resident comfort or energy usage [142]. It is stated that depending on the unique nation and area, incorporating PV power generation in buildings may offer approximately 20–75% of building energy usage, and greenhouse gas emissions would also be considerably reduced [143]. This is a new field for energy conservation and worldwide environmental preservation, and is being acknowledged by an increasing number of environmentalists. The incorporation of heat pipe PV&PV/T systems with BIPV/T technology exhibits a huge trend of future development and integration in this field. Due to the rarity of research studies on heat pipe PV systems used in building structures, many studies concentrate on the use of heat

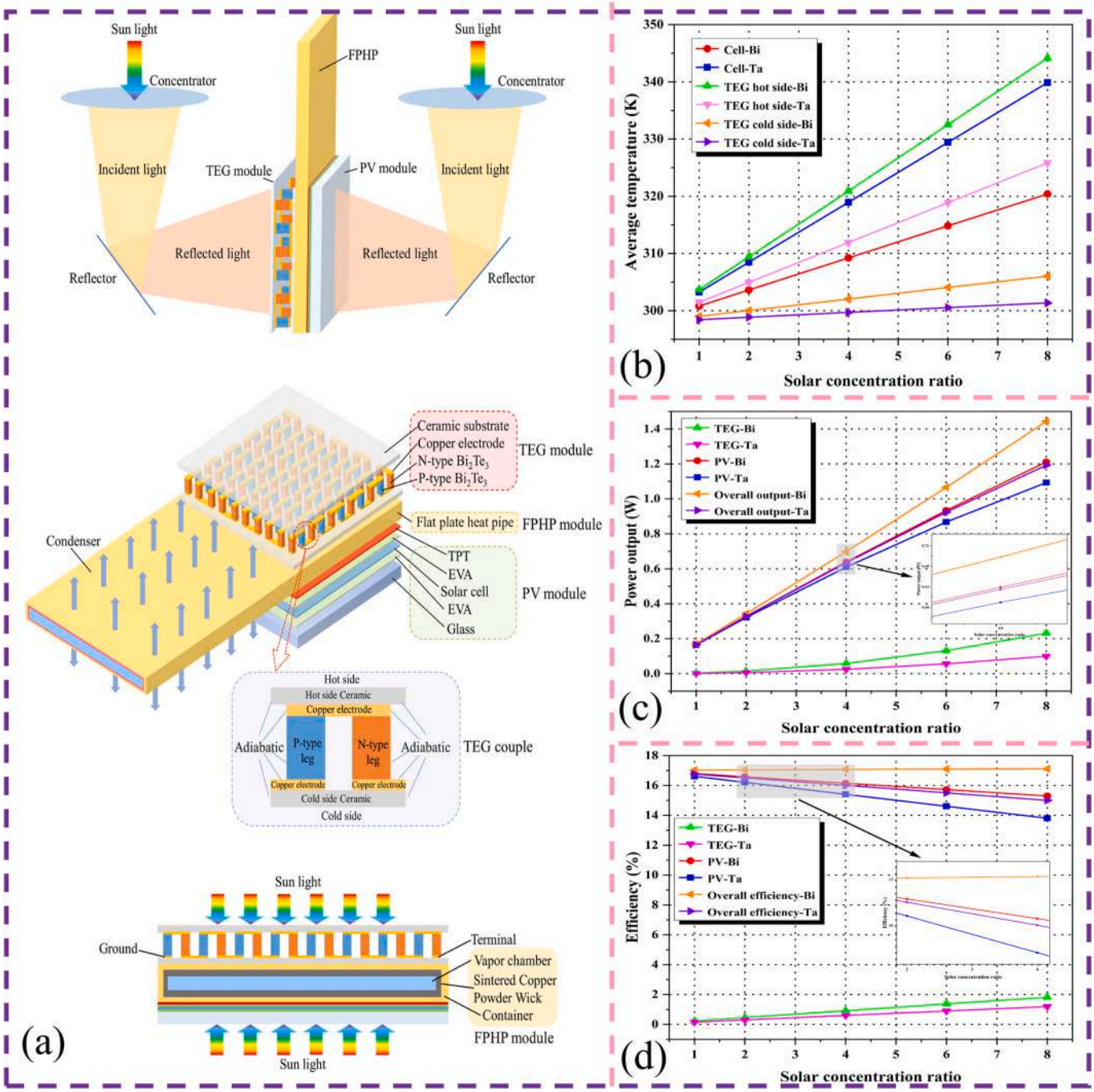



**Fig. 22.** Bifacial PV-STEG-heat pipe system (a) Schematic view, (b) Average temperature, (c) Power output, and (d) Electrical efficiency (adapted/reproduced with permission from Ref. [132]). Copyright © 2021 Elsevier.

pipe PV/T systems in connection with building structures. Therefore, the information included in this section focuses mostly on the heat pipe PV/T system in conjunction with building construction.

Wu et al. [29] established a BIPV-integrated heat pipe/PVT model to eliminate the uneven temperature uniformity of solar cells. The temperature uniformity of the PV cell was decent, and the temperature change was less than 2.5 °C, while the thermal and electrical efficiencies were 63.65% and 8.45%, respectively. Wang et al. [144,145] experimentally tested a BIPV/T system based on heat-pipe devices under indoor conditions (Fig. 25a). The principal body of the tested solar collector consisted of a PV panel, battery, charge controller, PCM (GR52), heat pipe, metal wires, and circulating water system. Xenon lamps were fixed at the front of the PV module to simulate the incoming sunlight. The PCM was used to store the additional heat released from the PV module and release energy to prolong the running hours. Added metal wires were prepared to enhance the low thermal conductivity of the PCM. As shown in Fig. 25b, after 8 h, the temperature of the tank water gradually increased to 314 K. Importantly, compared with the classical BIPV/T system, a significant enhancement of 25.5% in overall efficiency was observed.

Hui et al. [146] proposed a dynamic model to predict the performance of a BIPV/T system combined with a heat-pipe device. The calculated results revealed that the water heating and photoelectric efficiencies were 35% and 10%, respectively.

As shown in Fig. 26, Jouhara et al. [147,148] explored a novel structure that used a heat-pipe PV/T system as the building envelope. A

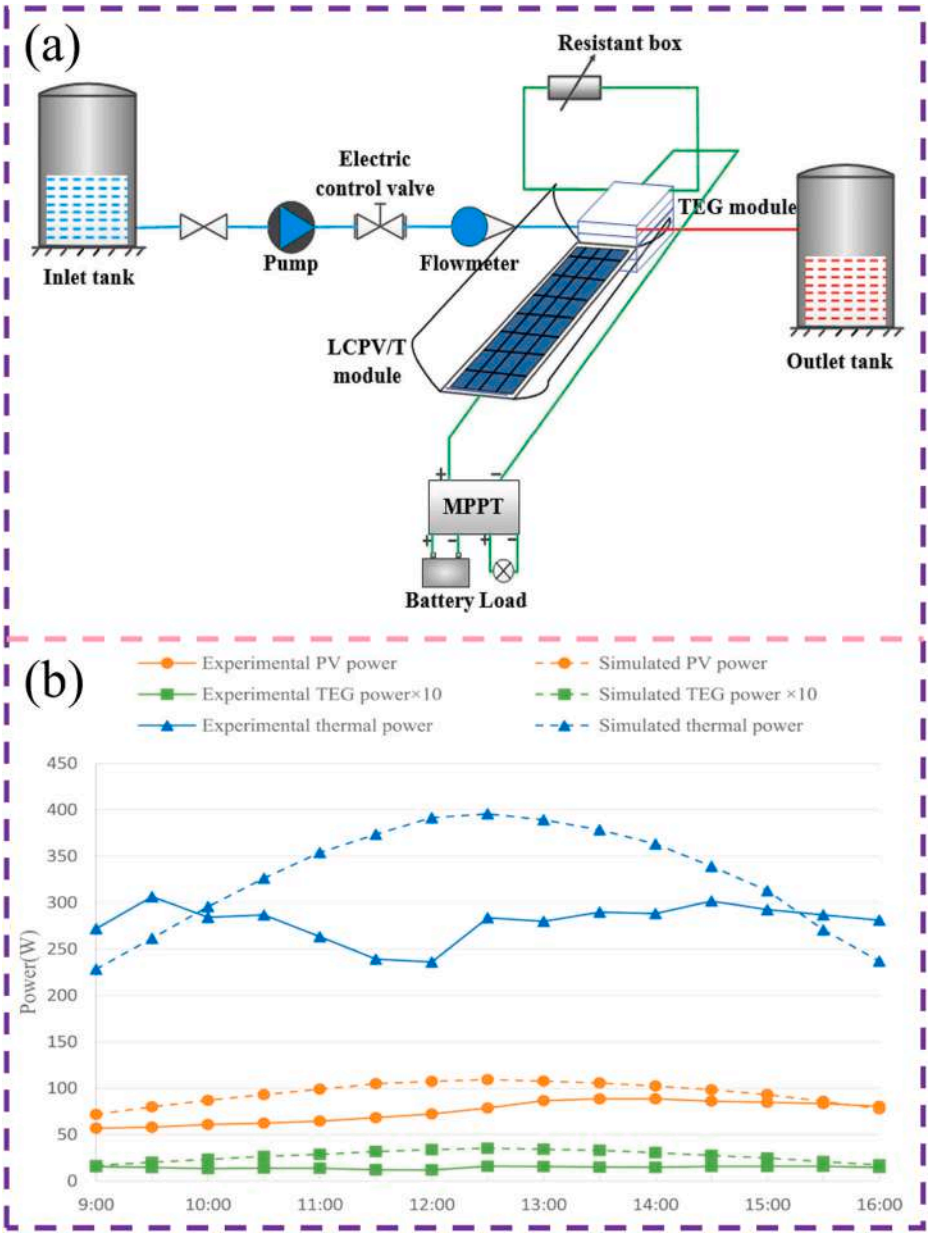



**Fig. 23.** Heat pipe-LCPV/T-TEG system (a) schematic picture of the layout, (b) (adapted/reproduced with permission from Ref. [135]). Copyright © 2019 Elsevier.

heat mat solar collector with a length of 400 cm and a width of 40 cm was combined with a PV panel to serve as the building envelope. They fabricated a full-size wood structure prototype using a heat mat as the envelope material to test the compatibility of the novel setup. The area of the envelope was classified into three parts: an area installed with a PV panel without cooling, a PV/T panel with heat mat cooling, and a heat mat thermal collector. The hybrid fluid of water/glycol was used as the intermediate fluid to remove heat from the heat mat. After an approximately clock field test, they found that the overall efficiency of this new layout was approximately 50%.

Liu et al. [149] inspected a ventilator system combined with a PV/T system using a heat pipe to heat the air during the cold season. The reported results indicate that this novel could provide not only fresh air but also had an average thermal efficiency of 26.7%. Moreover, Brinkley et al. [35] demonstrated an advanced low-cost and eco-friendly optical PV/T system that could be mounted on a rooftop or façade. The highlight of the proposed system was the unique layout of the PV/T collector. The author skilfully attached solar cells to both sides of the heat pipe and encapsulated them in a glass tube. The bottom of the tube was coated with a highly reflective material to reflect sunlight to the backside of the heat pipe. Each collector was composed of 26 solar cells connected in series and in parallel. It was projected that by implementing the newly developed PV/T system, roughly 4.3 tons/$m^2$ of carbon emissions could be avoided after operation for two decades. (Fig. 27).

At this time, the primary discrepancy is the mismatch between the building demand and the hot water and electricity produced by the heat pipe BIPV/T system. because intermittent solar radiation causes the power and hot water produced to be unstable. However, there are notable variations in building energy usage from peak to valley. Additionally, this technology is relatively costly, and more research is still needed to determine its long-term operating costs.

### 4.3. Combined with PCM technology

The term "phase change materials" refers to materials that change state and provide latent heat at a constant temperature. The process of

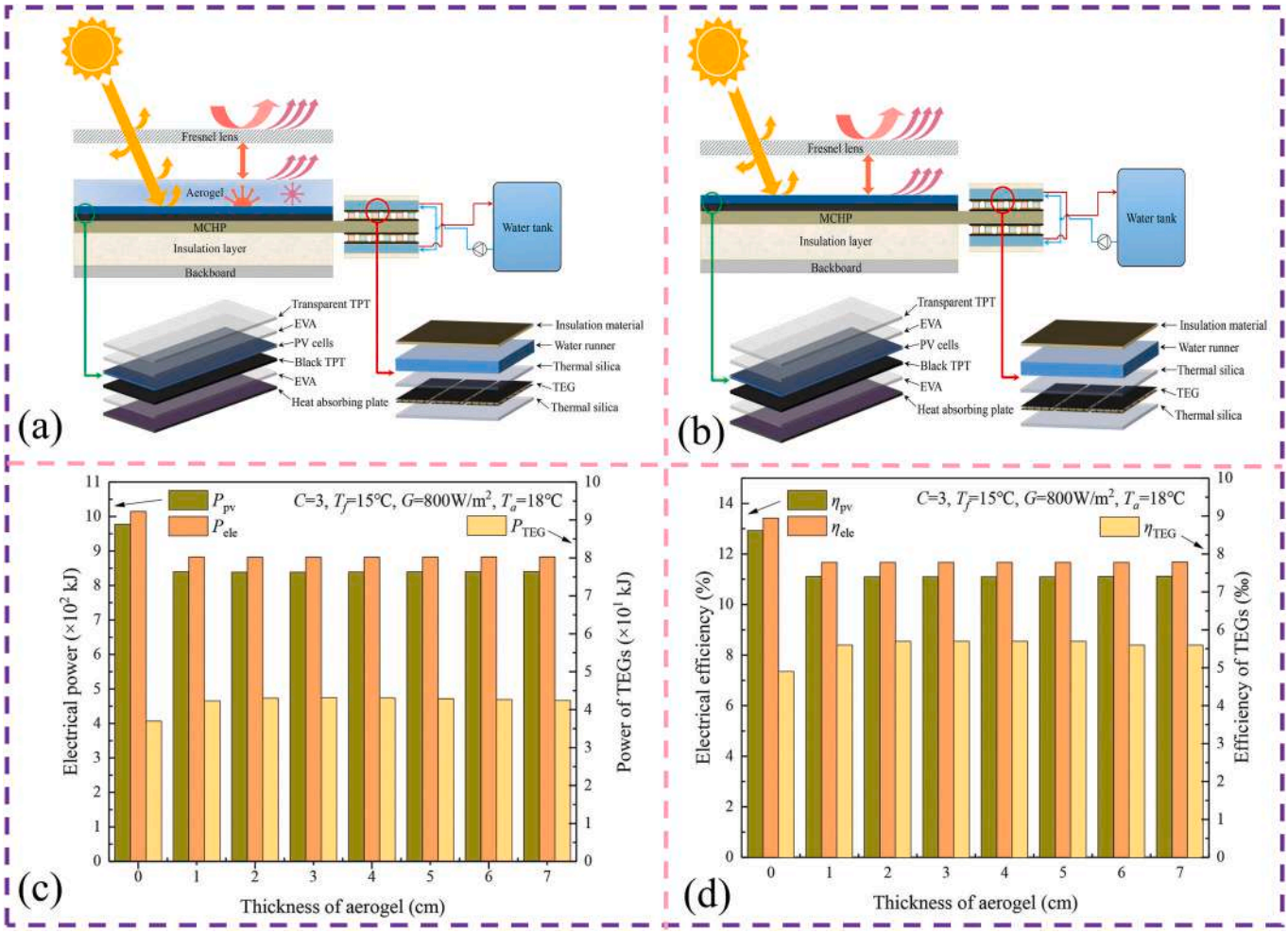



**Fig. 24.** Diagram of the LFPV/T-TEG system (a) with aerogel glazing, (b) without aerogel glazing, (c) effect of aerogel thickness on power generation, and (d) effect of aerogel thickness on conversion efficiency (adapted/reproduced with permission from Ref. [138]). Copyright © 2021 Elsevier.

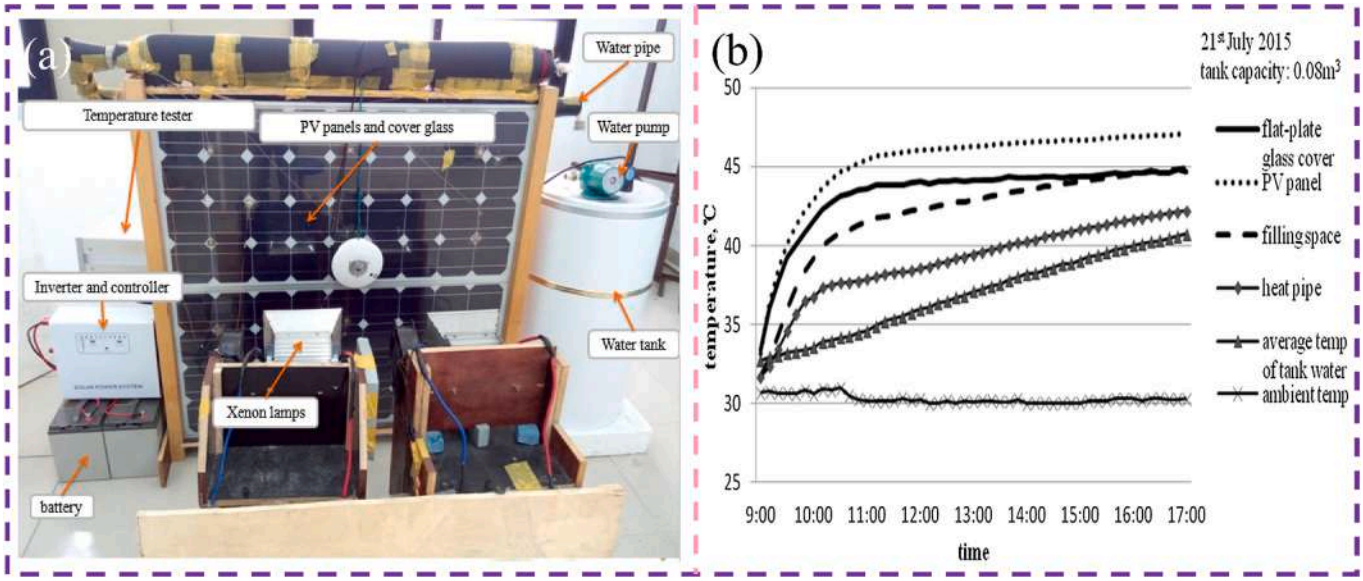



**Fig. 25.** Heat pipe-BIPV/T system (a) image of the test rig, (b) results of the temperature variation (adapted/reproduced with permission from Ref. [144]). Copyright © 2016 Elsevier.




**Fig. 26.** Image of the BIPV/T system (a) heat pipe device (b) building envelope fabricated with heat pipe PV/T system (adapted/reproduced with permission from Ref. [147]). Copyright © 2016 Elsevier.

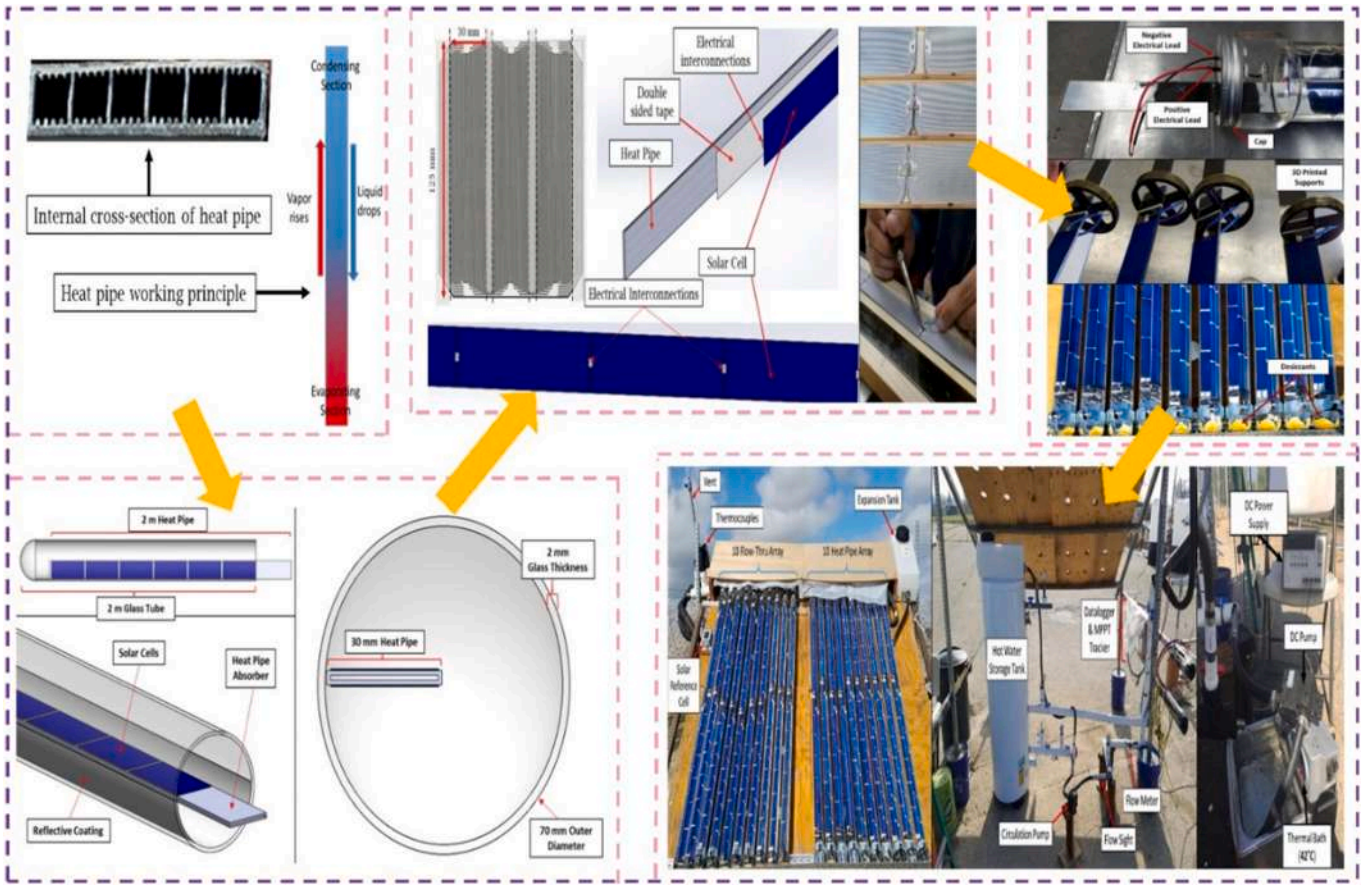



**Fig. 27.** Manufacturing process of the advanced optical PV/T system (adapted/reproduced with permission from Ref. [35]). Copyright © 2020 Elsevier.

altering the physical characteristics is known as the phase change process, and it involves the PCM absorbing or releasing a large amount of latent heat. Generally, PCM are classified in various ways based on several PCM characteristics. Typical categorisation methods are defined by the chemical properties and phase change mechanism, as listed in Table 3.

**Table 3**
Classification of PCM.

| Phase transition process | Chemical property | | |
|---|---|---|---|
| Solid-Liquid | Organic | Inorganic | Eutectic |
| Solid-Solid | Paraffins | Salts | Organic-Organic |
| Solid-Gas | Senior fatty acids | Hydrated salts | Organic- Inorganic |
| Liquid-Gas | Polyolefins | Molten salts | Inorganic-Inorganic |
| / | Alcohols | Mixed salts | / |

The most frequently used organic PCM are paraffins, senior fatty acids, polyolefins, and alcohols. The benefits of significant latent heat are that there is no super-cooling, no corrosion, and no volume impact. Importantly, the hazardous and harmless free has led many researchers to study organic PCM extensively in the context of solar PV systems [150–152]. Salts, hydrated salts, molten salts, and mixed salts are main examples of inorganic PCM that are often used as optional materials for latent heat storage. Generally, a single-component hydrated salt can no longer satisfy the demands of the growing phase change energy storage technology. Further studies have revealed that the addition of hydrated salts lowers the phase transition temperature of the system. Consequently, mixed salts are gaining popularity, and a distinct advantage of this form of PCM is that the melting temperature can be varied by varying the mixture of different salts [153]. Eutectics are blended with either organic or inorganic materials to create the desired PCM, and its melting temperature can be determined if the mass concentration of each component is altered [17].

The PCM technology applied in solar PV systems has been widely researched in recent years, and the most advanced achievements in the related issues described in the literature have been evaluated in Refs. [154–157]. In 1978, PCM was proposed as a thermal energy storage material with a PV panel. Despite the suggestion, the use of PCM for PV cooling was not investigated until the 1990s [17]. When PCM is utilised for PV panel thermal management, it is primarily employed as a passive cooling technique that does not require any additional equipment. Parasitic energy consumption is avoided, and reliability is increased. Additionally, the temperature of the PV panel may remain relatively constant, which is crucial for the operation of solar cells [158]. In terms of PV/T systems, combining PCM with PV/T technology has the potential to significantly enhance the energy conversion efficiency of a PV/T system. The heat collected by the PCM can be utilised to heat the room, provide hot water, and perform other daily tasks. A comparison of the performance differences of PV/T and PV/T-PCM systems revealed that incorporating a PCM in a PV/T system can clearly increase the energy performance of the system [159].

Özbaş [160] designed a novel passive cooling method by inserting TPCT into PCM block to cool down the solar cells. The author suggested that an increment of 6% can be obtained when methanol was selected as working medium. The 24 h performance of the heat pipe cooled PV-TEG system incorporating with PCM was evaluated in Ref. [161]. By releasing the heat absorbed during the day, the proposed layout could generate power at night to achieve higher electricity revenue. This work provides new thought on how to improve the all-day performance by introducing PCM into PV-TEG system. Sweidan et al. [162] proposed a transient model to evaluate a PCM-based heat pipe PV/T system under specific test conditions. Using the PV/T system to provide hot water was considered energy-saving and feasible, with a payback time of 13.7 years. The application of a phase-change slurry in a heat-pipe PV/T system was tested in Ref. [163]. Based on the experimental results, it can be confidently stated that the performance of the PV/T system using the phase-change slurry was significantly enhanced compared with the water-type PV/T system. Wang et al. [164] created a unique PV/T

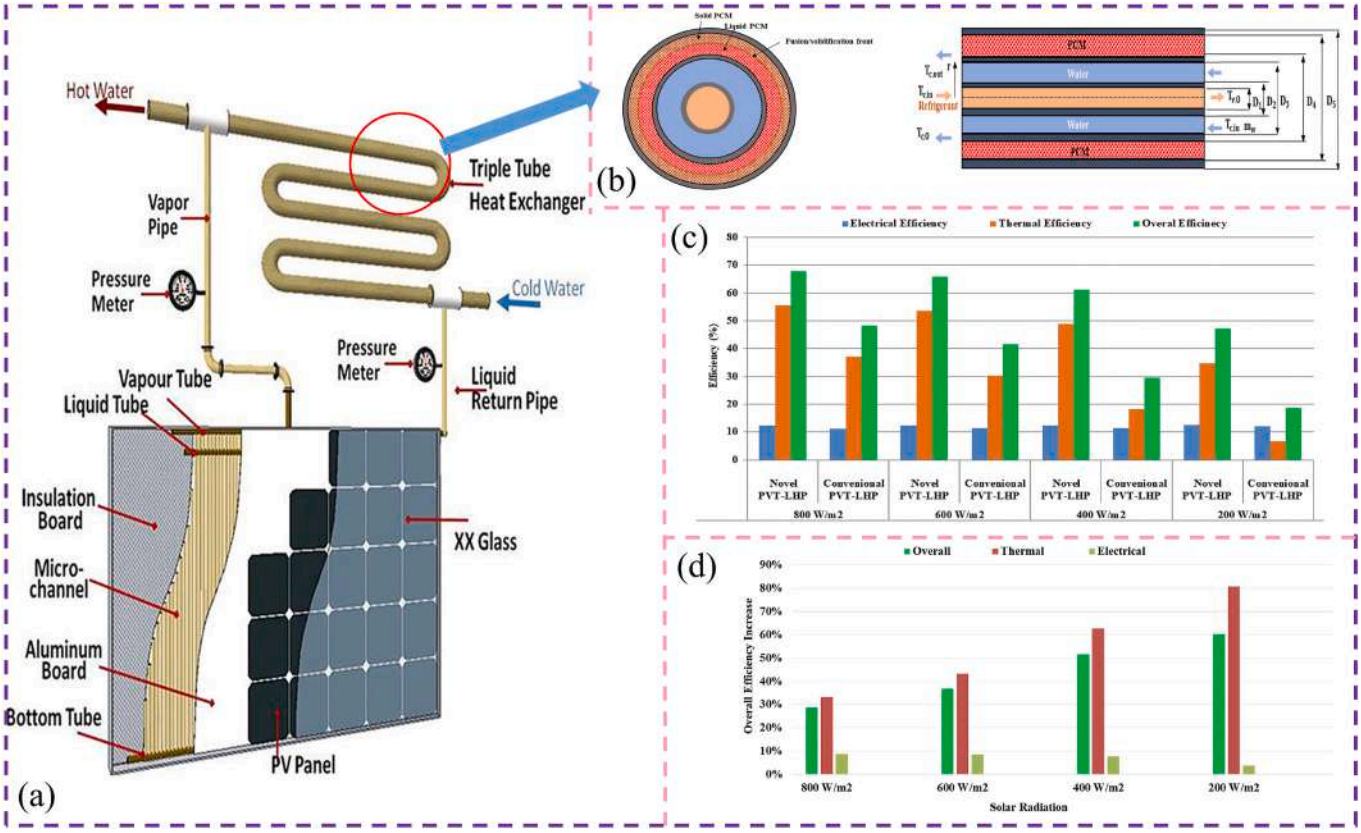



**Fig. 28.** LHP-PV/T system combined with PCM heat exchanger (a) schematic view of the overall system, (b) detailed structure of the PCM heat exchanger, (c) variation of energy efficiency, and (d) increase of total energy efficiency (adapted/reproduced with permission from Ref. [165]). Copyright © 2019 Elsevier.

system using solid-solid PCM. They achieved an average increase in total energy efficiency of 5–30%.

Diallo et al. [165] fabricated an LHP-PV/T system using a specially designed PCM heat exchanger (Fig. 28a) at the condenser section. In contrast to the traditional heat-pipe PV/T system, in this design, the PCM was not installed at the rear of the PV/T collector, and it was directly incorporated into the water pipe. As illustrated in Fig. 28b, a coaxial heat exchanger charged with PCM in the outer space was designed to collect the latent heat released from the refrigerant R-134a. A numerical model is developed to investigate this novel system. From Fig. 28c and **d**, it is evident that the thermal efficiency of the proposed system was 33% higher than the reference system, and the electrical efficiency also increased by 8.7%. In the study of Yu et al. [166], a similar setup was also built to test the LHP-PV/T system equipped with PCM heat exchanger. The height difference between evaporator and condenser part was detailly analyzed, and the optimal height difference of the current system was about 1.1 m.

A comparative assessment conducted by Modjinou et al. [167] further confirmed that the use of PCM could enhance the performance of the designed PV/T system. Besides, González-Peña et al. [33] found that using a heat pipe can enhance the heat transfer effect during the melting of the PCM, leading to a higher overall performance in a heat pipe PV/T system.

Economic cost is an essential consideration when using PCM technology, Gad et al. [168] performed an examination to find out the economic behaviour of PCM based heat pipe PV system. According to their results, the payback period and electricity cost of the PCM based heat pipe PV system was significantly lower than conventional pure PV system. In another study, Gad et al. [169] furtherly discovered that the power cost savings of photovoltaic/heat pipe-phase change material (PV/HP-PCM) system was 13.8% lower than the PV only system. These results indicate that using PCM in heat pipe PV system is economically feasible. However, the economic evidence of utilizing PCM in heat pipe PV/T system is still insufficient.

The majority of PCMs used in solar energy systems are organic, which has inevitably led to certain issues. This is one of the technology's drawbacks. For instance, corrosion problems often arise, especially when a PCM and a heat pipe device are enclosed in the same container. The cyclic stability of PCM should also get special consideration.

### 4.4. Combined with heat pump technology

Heat pump technology is an efficient and energy-saving technology that fully exploits low-grade heat energy [170]. It is a well-known natural rule that heat can spontaneously transmitted from a high-temperature body to a low-temperature one, but not vice versa. A heat pump is a mechanical equipment that uses reverse circulation to transfer heat from low-temperature sources to high-temperature sources. The cost is that a small amount of high-grade energy (commonly referred to as electricity) is consumed; however, the payoff is that a large amount of essential heat energy can be obtained. It may successfully use low-grade heat energy (i.e., from an air source, water source, or ground source) that is otherwise difficult to apply to satisfy the goal of energy savings.

In 1997, Ito et al. [171] suggested the first use of PV/T modules in conjunction with a heat-pump system. From the year when the initial notion was suggested to the present, systematic research has advanced significantly. One form of solar-aided heat pump system is a layout connected to a PV/T system (SAHP). Moreover, the system can be divided into two types: a direct expansion solar-assisted heat pump (DX-SAHP) and an indirect expansion sun-assisted heat pump (IX-SAHP) (IDX-SAHP). In an IDX-SAHP system or a conventional SAHP system, the PV/T collector and heat pump operate independently. A heat exchanger and heat transfer medium transport the heat collected by the PV/T collector to the heat pump unit. In contrast, the PV/T and heat pump systems are merged and integrated harmonically in the DX-SAHP system. The refrigerant in the pipe passing through the PV/T collector absorb solar heat to transform from liquid to gas employed as the evaporator of the heat pump system [170]. Utilizing the combined approach considerably enhanced the heat pump system performance by reducing the heat loss from the PV/T collector to the surrounding environment, and the structure of the direct expansion configuration further increased the heat pump COP by raising the temperature in the evaporator [172].

The multi-functions of SAHP systems, such as water heating, energy storage, space heating, and drying, have been extensively studied [173]. The PV/T-SAHP system may also be employed in a number of home applications, as well as in a variety of industrial activities that require low-temperature heat. Many of these processes can be found in the food and beverage, textile, mechanical, and pharmaceutical sectors [174].

Zhang et al. [176] developed a novel LHP PV/T system combined with a heat pump system to supply hot water. The results indicated that the COP of the system can reach up to 8.7, and the authors also discussed the influence of different parameters. Li et al. [177] developed a mathematical model for a solar/air or air-source heat pump (ASHP) LHP heat pump PV/T system. The model was validated using an actual test rig, and the lifespan cost of the proposed system was reduced by 29.6%. Furthermore, Li et al. [175] conducted an optimization study to obtain the optimal working conditions for the hybrid system. The obtained results indicated that with a tank volume of 150 L, the energy consumption of the system could be reduced by 55.7% (Fig. 29). The most recent advances and highlights of this research are presented in Table 4.

The contact area of a heat pump based on a round shape is very small to deliver the heat compared to a flat shape, which may affect the performance of the heat pump even if this type of technology has greater performance. Furthermore, the heat input of an evaporator changes significantly, making it impossible to ensure the stability of its functioning due to the unequal distribution of solar energy resources over time and location. The lengthy payback period and hefty initial cost also prevent its scaling up.

### 4.5. Combined with nanotechnology

Nanotechnology is a state-of-the-art technique for exploring the properties and applications of materials with structural dimensions in the range of 1 nm–100 nm [184]. At this scale, nanomaterials exhibit distinct physical and chemical properties and effects, such as surface effects, small size effects, quantum size effects, and macroscopic quantum tunnelling effects, which allow nanoparticles to differ from ordinary particles in terms of thermal, magnetic, optical, sensitive, and surface stability. The utilisation of nanoparticles in the field of improved mass and heat transfer is an extension of the existing high-efficiency heat transfer technology, and nanofluid is the result of this study. The modified nanofluid with tiny particles and a large specific surface area outperformed the base fluid in terms of heat transfer performance.

By scattering various nanoparticles into a basic liquid, a nanofluid can be created. The nanoparticles disseminated can be Fe, Al, Cu, Ag, Au, Ni (metal-based), $Fe_2O_3$, $Al_2O_3$, ZnO, $Fe_3O_4$, $TiO_2$, $SiO_2$, CuO, and SiC (nonmetal-based). Furthermore, carbon-based nanoparticles (carbon nanotubes, multi-walled carbon nanotubes, and single-walled carbon nanotubes) have been widely discussed. Water and ethylene glycol are the most widely used base fluids [185].

The van der Waals force between the nanoparticles causes the particles to adhere strongly. Clusters are formed when a significant number of nanoparticles attach to one another. These clusters grow and lure neighbouring clusters to merge with them. When they reach a specific level of development, there are more visible settlement phenomena [187]. Therefore, preparing long-term, stable, and evenly dispersed nanofluids remains challenging. The most prevalent methods for preparing nanofluids are one-step and two-step procedures [186]. Fig. 30 shows the preparation of the one-step and two-step procedures. During the preparation of nanoparticles, the one-step approach involves

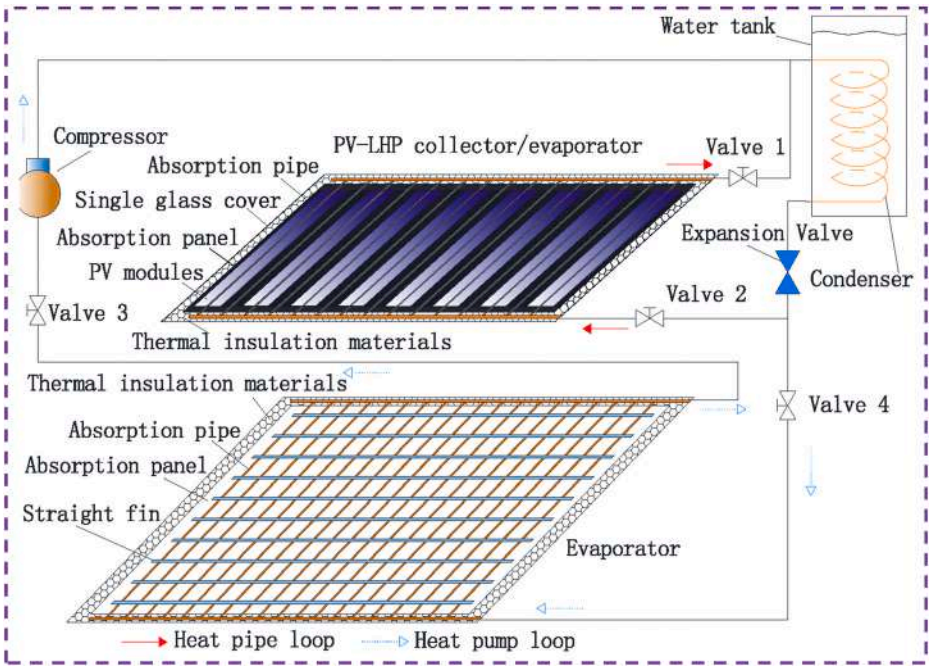



**Fig. 29.** Diagram of a LHP-PV/T-SAHP system (adapted/reproduced with permission from Ref. [175]). Copyright © 2022 Elsevier.

immediate dispersion of the particles into the base solution. Nanoparticles and nanofluids were created simultaneously. The two primary approaches for producing nanoparticles in a single step are gas- and liquid-phase methods. The two-step approach first involves preparing nanoparticles before dispersing them into the base solution to generate nanofluids [186]. Compared to the one-step procedure, the two-step method was more suitable for laboratory size preparation. Magnetic stirring and sonication time are frequently utilised as supplementary preparation methods.

At present, research studies involving nanotechnology within the area of heat pipe PV and PV/T systems have seldom been reported. Previously published articles originated from different directions; some authors used nanoparticles with the base fluid as the optical spectrum beam splitter (SBS), while others have attempted to utilise the hybrid nanofluid as the working medium of the heat pipe. As illustrated in Fig. 31a, Han et al. [188] provided a dynamic model for predicting the performance of an SBS PV/T system. Accordingly, a mixture of Ag/CoSO4 nanoparticles was dispersed into the base fluid of propylene glycol as an optical SBS, and the heat pipe was joined to the back layer of the solar cell. SBS and heat pipe devices were used as the first and second thermal absorbers, respectively. The nanofluid-filled SBS was placed above the PV panel to filter out useless energy in the solar spectrum. As presented in Fig. 31b and **c**, the findings reveal the interaction between the coolant flow rates and concentration ratios on the output performance. Under high solar radiation conditions, it was beneficial to improve the thermal efficiency by reducing the mass flow rate of the nanofluid. Moradgholi [36] experimentally investigated the performance of a TPCT PV/T system using methanol and aluminium oxide/-methanol as the working medium. Accordingly, an optimal filling volume of 50% and nanomaterial concentration of 1.5 wt% was suggested. The Enhancements of average photoelectric conversion and overall efficiencies of 1.0% and 27.3% were achieved, respectively. Sami et al. [189] developed a mass and heat transfer model to study the effect of using a magnetic nanofluid in a heat-pipe PV/T system. Based on a numerical study, the authors declared that a higher magnetic field energy could lead to a stronger heat transfer performance.

Furthermore, Samuel Sami [189] designed an indirect heat-exchange PV/T system, and a heat pipe was utilised as the heat exchanger to transport the thermal energy from the PV/T module to the water tank. The thermal absorber was adhered to the rear of the PV panel, different nanofluids (such as CuO, $Al_2O_3$, $Fe_3O_4$, and $SiO_2$) were considered as the heat transfer fluid, and a magnetic field generator was bonded with the pipeline to explore the influence of the magnetic field. Based on the reported data, the use of CuO nanofluid exhibited the best results, and a higher concentration of nanofluid resulted in higher system performance. In addition, the overall thermal and total output performances of the PV/T system can be enhanced by adopting a stronger magnetic field.

Some disadvantages of this technology can be summarized as: the key to using nanotechnology effectively is the long-term stability of the nanofluids. Additionally, if the nanofluids function as an optical filter, the optical loss brought on by the scattering of incident light is another issue that must be taken into consideration.

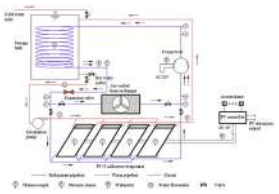
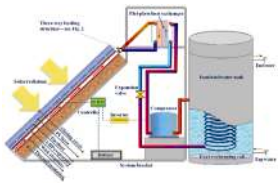
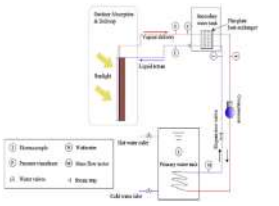
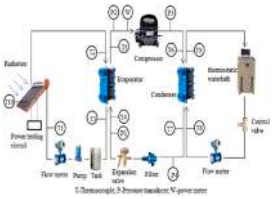
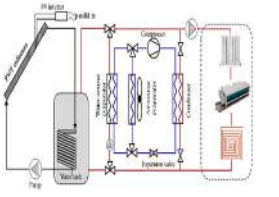
In summary, detailed results and observations of the hybrid systems incorporating the developed novel technologies are presented in Table 5.

## 5. Challenges and future suggestions

As a passive and efficient thermal management technology, the application of heat pipe devices in PV&PV/T systems has provided new solutions to address existing issues and has gained increasing interest over the past years, particularly in the fields of innovative technologies. Turning to challenges and future suggestions, some plausible directions and perspectives based on this review are identified as follows.

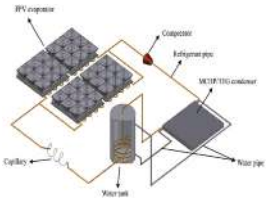
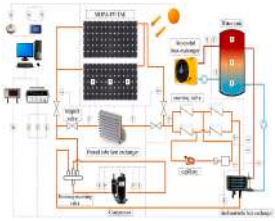
- Although crystalline silicon cells are extensively used in CPV systems, multijunction cells with higher energy conversion efficiency and low temperature sensitivity are recommended for medium-to-high CPV systems.
- Currently, PHP-based systems primarily focus on miniaturisation, and research studies have rarely been conducted. To fulfil the demand for solar PV systems, it is imperative to construct large-scale PHP with innovative exterior structures and analyse their heat transfer limit, heat transfer distance, and functioning characteristics, posing a new challenge to PHP theory and heat transfer improvement research.
- For heat pipe PV system, the waste heat released from the condenser part could be recycled again instead of rejecting in vain. For instance,

**Table 4**
Summary of the heat pipe-based PV/T systems integrated with heat pump technology.

| References | Region | Methods | Schematic view of the system | Combination mode | Types of heat pipe | COP | Highlights |
|---|---|---|---|---|---|---|---|
| [178] | Hong Kong, China | Experimental |  (adapted/reproduced with permission from Ref. [167]). Copyright © 2012 Elsevier. | Direct-expansion | CTHP | 3.32–4.01 | ● The heat pipe was employed in the heat pump PV/T system.<br>● The system can switch between three operation modes.<br>● Both energy and exergy analyses were conducted. |
| [179] | Hull, UK | Simulation |  (adapted/reproduced with permission from Ref. [168]). Copyright © 2014 Elsevier. | Indirect-expansion | LHP | 5.51: only thermal performance<br>8.71: both thermal and electrical performance | ● LHP with vapour-liquid separation setup was introduced in heat pump PV/T system.<br>● The performance of the system was analyzed from a socio-economic aspect.<br>● The COP of the proposed system was 1.5–4 times higher than the conventional heat pump system. |
| [180] | Hefei, China | Experimental& Simulation |  (adapted/reproduced with permission from Ref. [169]). Copyright © 2015 Elsevier. | Indirect-expansion | LHP | 4.93: average value | ● A novel LHP heat pump PV/T system was applied for building heating.<br>● The thermal efficiency of 22.2% was observed by integrating with the heat pump system. |
| [181] | Beijing, China | Experimental& Simulation |  (adapted/reproduced with permission from Ref. [170]). Copyright © 2017 Elsevier. | Indirect-expansion | CTHP | 4.08–4.26: only thermal performance<br>5.10–5.43: both thermal and electrical performance | ● The influences of various parameters (input solar energy, inlet temperature, pitch space of heat pipe, absorptivity of plate, ambient temperature) were discussed. |
| [182] | Nanjing, China | Simulation |  (adapted/reproduced with permission from Ref. [171]). Copyright © 2019 Elsevier. | Indirect-expansion | TPCT | 2.43–4.13 | ● The system can be shifted between solar or air source operational modes.<br>● The energy consumption can be reduced by 13.1%, compared with a single ASHP system.<br>● The accumulated electricity increased by 14.7, compared with a single PV system. |

(*continued on next page*)

**Table 4** (*continued*)

| References | Region | Methods | Schematic view of the system | Combination mode | Types of heat pipe | COP | Highlights |
|---|---|---|---|---|---|---|---|
| [34] | Hefei, China | Simulation |  (adapted/reproduced with permission from Ref. [32]). Copyright © 2021 Elsevier. | Direct-expansion | MHPA | 4.73–8.62 | ● A newly designed layout combined with MCHP/TEG devices and PV/T panel was proposed.<br>● The performance of the hybrid system can be significantly enhanced by applying of TEG unit. |
| [183] | Beijing, China | Experimental |  (adapted/reproduced with permission from Ref. [172]). Copyright © 2021 Elsevier. | Direct-expansion | MHPA | 2.8: solar mode<br>2.1: air and solar mode<br>2: air mode | ● The proposed system's optimal yearly operating plan was offered.<br>● The PV/T panel was evenly cooled. |

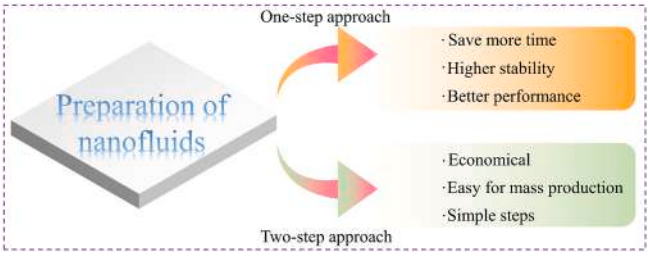



**Fig. 30.** Schematic of nanofluid preparation [186].

when paired with a solar still, the extra heat might be utilised to warm up the sea water to hasten the desalination process. Although there have been attempts in this area, they are still far from sufficient.

- One-dimensional mathematical models serve as the foundation for many models and simulation results. One of the most significant drawbacks is that the overall performance characteristics of the system cannot be established and some departure from the actual scenario may occur. Consequently, three-dimensional model development is strongly advised for analysing the application of heat pipes and other innovative technologies in solar PV systems (if feasible).
- Owing to the nature of synergistic energy production, the use of TE devices in heat pipe solar PV and PVT systems is both accepted and encouraged. However, numerous researchers have focused on undertaking theoretical modelling and simulations, and experimental work needs to be expanded further. Moreover, data from actual situations are useful for guiding the growth of this discipline. Therefore, practical experiments conducted under transient settings are strongly recommended. In addition, a novel structural design is encouraged to obtain a superior overall performance (such as the developing layout of the sandwich arrangement, which allows higher power generation).
- The integration of PCM in the heat-pipe PV/T system could result in a higher thermal performance. Despite this, the high initial cost, lower thermal conductivity, and flow phenomena after melting lead to a longer payback period, poor heat-transfer performance, and sealing issues. Perhaps, the use of form-stable PCM could overcome this obstacle, while preliminary experimental investigations are lacking. Moreover, considering the heat transfer enhancement of nanoparticles and metal foams is an upgraded option. Furthermore, owing to the sporadic and unpredictable nature of solar resources, an energy storage system that employs PCM as an energy storage carrier is a better choice for pairing with a heat-pipe PV/T system. However, a long-term economic study is required to ensure that the proposed system is feasible, the research on this topic is not enough.
- It has been demonstrated that using BIPV technology in a heat-pipe PV/T system is useful for energy savings in green buildings. However, it remains limited to laboratory settings rather than in real installations in buildings. Additionally, the incorporation of various energy storage and conversion systems (such as thermal energy storage, ventilation devices, and heat pumps) is recommended. The significance of architectural aesthetics cannot be overstated, necessitating more delicate and suitable integration methods.
- The combination of heat pump technologies has gained considerable attention, and many studies have been conducted using numerical and short-term experiments. In most cases, a dual-source heat pump outperforms a single-source heat pump because it can simultaneously fulfil the heating, cooling, and hot water demands. Furthermore, the performance of the heat pump may vary depending on meteorological conditions. However, there is a severe shortage of long-term operating data for real systems. Besides, the higher initial cost of the device should be considered from the economic feasibility.
- Nanotechnology has been widely employed to improve heat transfer. Hence, nanoparticles are always present in the working fluids and other materials (such as PCM). To date, the use of nanotechnology has seldom been examined, whether through simulations or experiments, and the research gap in this sector has to be filled. For example, the effects of incorporating different nanoparticles into the base fluid, PCM, and heat pipes, have rarely been described. Furthermore, nano treatment for the heat transfer device is advised to improve the heat transfer. Moreover, the establishment of optical mathematical model is critical to understand the optical loss of the incident sunlight when nanofluid is used as filter.

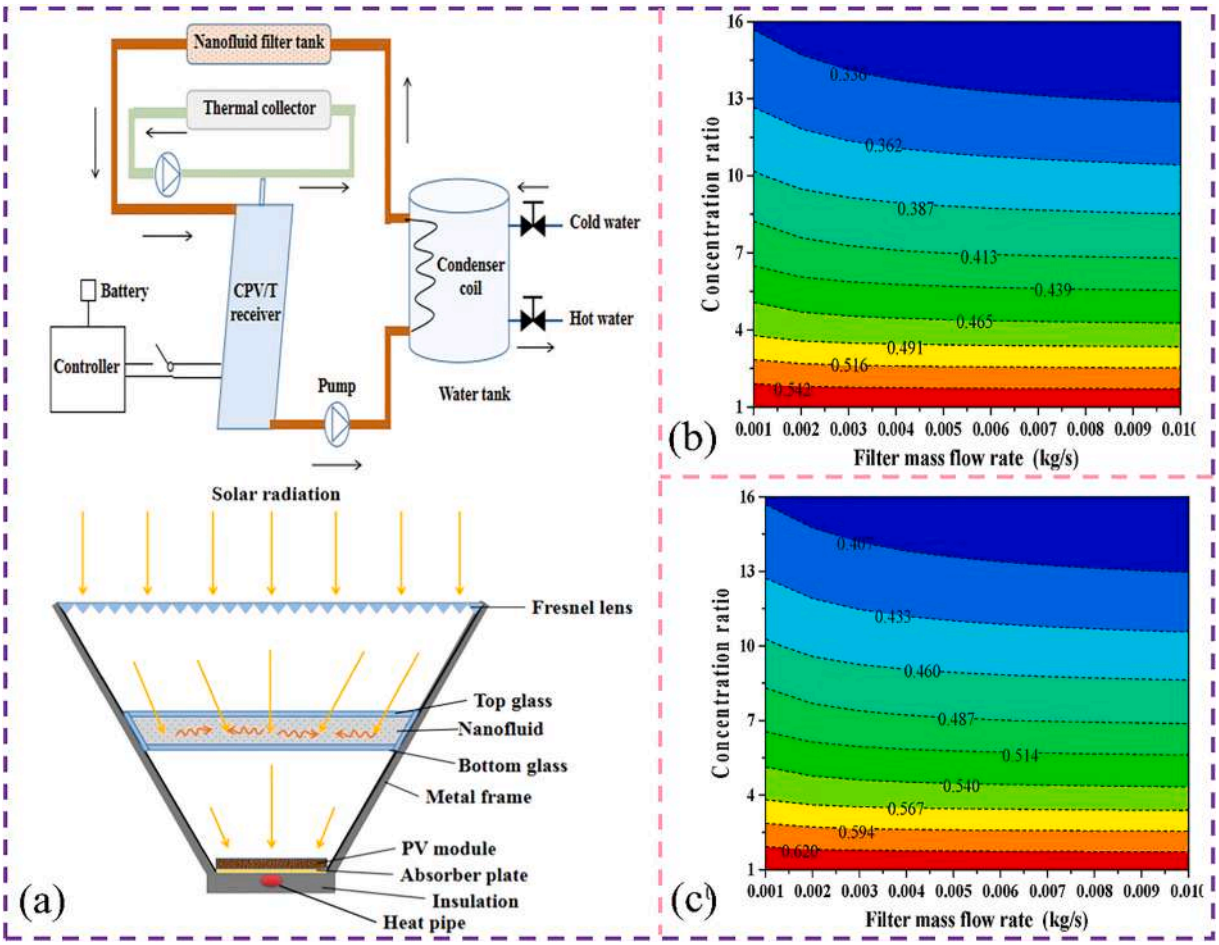



**Fig. 31.** Nanofluid cooled heat pipe PV/T system (a) diagram of the experimental setup, (b) average thermal efficiency, (c) average total efficiency (adapted/reproduced with permission from Ref. [188].) Copyright © 2020 Elsevier.

- The employment of a single technology can no longer satisfy the system's total set of requirements, and the usage of several technologies is the way of the future in terms of research and development. There have been new advances in this area, for example, the combination of TE and PCM technologies to realize all day power generation.
- To evaluate the performance of various types of heat pipe PV&PV/T systems from different perspectives, a thorough evaluation method or model should be devised. Future works are actively encouraged in terms of cleanliness and sustainability, as well as economic and social aspects, particularly in terms of environmental effects and socio-economic factors. When this is completed, it is easy to compare various setups simultaneously.
- Further study is required to determine how the usage of heat pipe PV&PV/T systems would affect carbon emissions in manufacturing and operation against the backdrop of global carbon neutrality (including environmental and exergoenvironmental analysis).

## 6. Conclusions

In this paper, a comprehensive review of the advancement of heat pipe PV&PV/T systems regarding innovative technologies is presented. Several heat pipes (particularly, TPCT, CTHP, LHP, MHPA, and PHP) can be joined with PV&PV/T systems to realize thermal regulation and higher energy conversion efficiency. Meanwhile, for comprehensive energy utilisation, various innovative technologies (such as TE, BIPV, PCM, heat pumps, and nano) can be integrated with PV&PV and PV/T systems. Several key points are summarized below based on the survey results.

- The developed PV&PV/T systems combined with various heat pipes (including TPCT, CTHP, LHP, MHPA, and PHP) have become a prominent research field owing to their significant heat transfer effects and energy conversion efficiencies. In addition, among the discussed hybrid layouts, the utilisation of PHP in PV&PV/T configurations is not as often used as other heat pipes.
- The application of a flat-plate-type heat pipe in conjunction with a PV panel is more appropriate and achieves a higher performance without using an extra thermal absorber plate. The directly connected structure has lower thermal resistance and is much easier to combine with other devices.
- Using heat-pipe cooling, a superior even temperature distribution of less than 2 K can be obtained. The cooling demand of the PV systems under 800X solar radiation can be satisfied with the assistance of a heat pipe.
- In cold regions or seasons, using a heat-pipe PV/T system with a proper operating fluid could eliminate the freezing and corrosion problems that hinder conventional water-type PV/T systems.
- Several parameters can affect the performance of a heat pipe PV/T system, including the addition of a glass cover, pitch of the heat pipe, filling ratio of the heat pipe, working medium of the heat pipe, inclination angle, tank capacity, and type of heat pipe.
- Integration with TE Technology. To maintain the large temperature gradient of the TEG module, it is recommended to work under concentrating conditions for heat pipe PV&PV/T systems with TE

**Table 5**
Summary of the combined technologies.

| Author(s) | Year | Methods | Heat pipe types | Combined technologies | | | | | Concentrated | Novelty | Critical conclusions |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | TE | BIPV | PCM | Heat pump | Nano | | | |
| Wu et al. [29] | 2011 | Theoretical | CTHP | | ✓ | | | | No | Heat pipe was introduced to the BIPV/T system to overcome the uneven temperature distribution of the PV panel. | ● The thermal and electrical efficiencies of the studied system were 63.65%, 8.45%, respectively. |
| Fu et al. [178] | 2012 | Experimental | Not mentioned | | | | ✓ | | No | A real design of the solar-assisted heat pump PV/T system was established. | ● The first and second laws of thermodynamic efficiencies were 36.5–38.4% and 7.4–7.8%. |
| Zhang et al. [176] | 2013 | Experimental& theoretical | LHP | | | | ✓ | | No | A novel LHP based PV/T heat pump system was developed for hot water supply and electrical generation. | ● The COP of the proposed system was 3.7.<br>● The electrical and thermal efficiencies were about 10% and 40%. |
| He et al. [134] | 2014 | Experimental& theoretical | Not mentioned | ✓ | | | | | No | A hybrid heat pipe PV/T system was established to drive the TE device to work in winter and summer mode. | ● The COP of the TE module, the electrical and thermal efficiency of the system are 1.7, 16.7% and 23.5%, respectively. |
| Zhang et al. [190] | 2014 | Experimental& theoretical | LHP | | | | ✓ | | No | The PV/T heat pump system combined with LHP was analyzed from the socio-economic aspect. | ● The payback time of the examined system in London, Shanghai and Hong Kong. |
| Wang et al. [144] | 2016 | Experimental | MHPA | | ✓ | ✓ | | | No | A new BIPV/T system charged with PCM using metal fillers to enhance the thermal conductivity was investigated. | ● Under the given conditions, the average efficiencies of 61.1%, 7.8% and 68.9% were obtained for the thermal, electrical and total. |
| Sweidan et al. [162] | 2016 | Theoretical | CTHP | | | ✓ | | | No | An optimization approach corresponding to the PV/T system with an energy storage feature was developed. | ● The payback time of the considered system was found to be 13.7 years. |
| Li et al. [128] | 2016 | Theoretical | MHPA | ✓ | | | | | No | The cost analysis of the PV-TEG system was conducted through the yearly power output. | ● The PV-TEG system integrated with MHPA is superior to that of a normal PV-TEG system in terms of economic and electrical performance. |
| Makki et al. [133] | 2016 | Theoretical | CTHP | ✓ | | | | | No | A theoretical model was presented to describe the impact of various ambient conditions on PV-TEG system. | ● The hybrid system is more suitable for working in some dry and hot areas (like the Middle East). |
| Hui et al. [146] | 2017 | Experimental& theoretical | TPCT | | ✓ | | | | No | Annual operation data of the BIPV/T system located in Hong Kong was analyzed. | ● The energy saving potential could reach up to 315 kW h/year/$m^2$. |
| Liu et al. [149] | 2018 | Experimental | CTHP | ✓ | ✓ | | | | No | The TE module connected with ventilator setup in a BIPV/T system was studied. | ● The system could drive fresh air into the room with an average COP of 6.4 and thermal efficiency of 26.7%. |
| Wang et al. [164] | 2018 | Experimental | MHPA | | ✓ | ✓ | | | No | The performance of the BIPV/T system combined with solid-solid PCM was thoroughly examined. | ● The total efficiency of the system could be greatly improved by using solid-solid PCM (5–30%). |
| Luo et al. [191] | 2018 | Experimental& theoretical | Not mentioned | ✓ | ✓ | | | | No | A novel BIPV-PV-TEG system was designed for meeting the cooling demand of hot areas in China. | ● The system has great potential for energy savings, with the values of 29.19–62.94 kW/$m^2$ compared to the conventional structure. |
| Chen et al. [163] | 2019 | Experimental | TPCT | | | ✓ | | | No | A new kind of working medium (phase change slurry) was selected and studied. | ● By adopting the new fluid, the thermal and electrical behavior of the system greatly improved. |

(*continued on next page*)

**Table 5** (*continued*)

| Author(s) | Year | Methods | Heat pipe types | Combined technologies | | | | | Concentrated | Novelty | Critical conclusions |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | TE | BIPV | PCM | Heat pump | Nano | | | |
| | | | | | | | | | | | ● 30% alkyl hydrocarbon material is considered as the working fluid. |
| Shittu et al. [131] | 2019 | Theoretical | MHPA | ✓ | | | | | Yes | A comparative study of PV-TEG and MHPA-PV-TEG systems were detailed discussed. | ● A 3D numerical model was prepared to describe the temperature and voltage distribution.<br>● An increase of 1.47% in efficiency was obtained compared to the PV-TEG system without heat pipe cooling. |
| Chen et al. [135] | 2019 | Experimental | MHPA | ✓ | | | | | Yes | The real field test of an LCPV/T-TEG system was presented and analyzed. | ● The overall efficiency of the LCPV/T-TEG system for one day test was about 57.03%. |
| Diallo et al. [165] | 2019 | Theoretical | LHP | | | ✓ | | | No | The LHP PV/T system equipped with a PCM heat exchanger was studied considering various parameters. | ● A 28% increase in total energy efficiency was found in comparison with the normal system. |
| Li et al. [129] | 2019 | Experimental | MHPA | ✓ | | | | | No | A preliminary study of the MHPA-PV-TEG system under summer conditions was performed. | ● The technical point of the MHPA-PV-TEG system was proved to be feasible. |
| Modjinou et al. [167] | 2019 | Experimental& theoretical | MHPA | | | ✓ | | | No | A comparative study of different PV/T systems (PCM, MHPA, conventional) was conducted. | ● For PCM, MHPA and conventional systems, the overall efficiencies were 36.71%, 35.53% and 31.78%, respectively. |
| Birol Kılkış [192] | 2020 | Experimental& theoretical | PHP | ✓ | ✓ | ✓ | | | No | Comprehensive research of PV/T system involved in PHP, TE and PCM was presented. | ● A prototype of using the novel technologies in a composite PV/T design was provided. |
| Shittu et al. [130] | 2020 | Experimental | MHPA | ✓ | | | | | No | Exergy analysis was performed to evaluate the performance of MHPA-PV-TEG system cooled by a water box. | ● The variation of the thermal efficiency of the proposed system was between 56.57% and 69.53%. |
| Han et al. [188] | 2020 | Theoretical | CTHP | | | | | ✓ | Yes | A dynamic model was proposed to define the heat pipe PV/T system combined with the nanofluid optical splitter. | ● The average total efficiency of the system was 53.66%, with the highest value of 73.20%. |
| Samuel Sami [189] | 2021 | Theoretical | TPCT | | | | | ✓ | No | A two-dimensional model was prepared to describe the influence of magnetic nanofluid on the performance of the PV/T system. | ● The CuO nanofluid led to the highest overall performance and heat pipe efficiency. |
| Wen et al. [137] | 2021 | Theoretical | TPCT | ✓ | | | | | Yes | The layout of the PV-TEG system using TPCT and water cooling was simultaneously studied. | ● The system cooled by TPCT could provide higher overall performance compared to the water-cooled layout. |
| Wen et al. [138] | 2021 | Theoretical | MHPA | ✓ | | | | | Yes | The effect of using aerogel material in PV-TEG system under various test conditions was explored. | ● The best performance of the system was achieved with the aerogel thickness of 30 mm. |

technology. Meanwhile, long-term real field examination is required to identify the economic feasibility.

- Integration with BIPV Technology. Owing to the compact structure and passive operation features, the integration of heat pipe PV&PV/T systems is suitable for satisfying the demand for building heating and electricity consumption. Moreover, the layouts of the hybrid system can serve as a part of the building structure. It is beneficial for building energy conservation, particularly in light of global carbon emissions reductions.
- Integration with the PCM technology. The majority of the considered materials are organic PCM based on the solid-to-liquid transition. An enhancement of the overall efficiency up to 30% could be accomplished in heat pipe PV/T system with PCM, while an increase of 33% and 8.7% in thermal and electrical efficiencies could also be found. In addition, owing to its low thermal conductivity, the melting progress can be enhanced by using heat pipes or metal additives.
- Integration with heat-pump technology. Different design strategies result in different levels of performance. Generally, depending on the

test conditions and operation modes, the COP of the tested system can be varied between 2 and 8.71.

- Integration with nanotechnology. As a crucial way of improving heat transfer, nanotechnology in heat pipe PV&PV/T systems has not reached its full potential. Hence, the research potential is enormous. There are still several gaps to be addressed (e.g., application forms of nanotechnology, the influence of different nanoparticles, and the added concentration of nanoparticles).

## CRediT authorship contribution statement

**Yuanzhi Gao:** Methodology, Investigation, Writing – original draft, Writing – review & editing. **Dongxu Wu:** Resources, Visualization. **Zhaofeng Dai:** Formal analysis, Methodology. **Changling Wang:** Conceptualization, Methodology. **Bo Chen:** Investigation, Formal analysis. **Xiaosong Zhang:** Supervision, Funding acquisition, Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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